Linköping University | Department of Computer and Information Science
Master Thesis | Cognitive Science
Spring Term 2026 | LIU-IDA/KOGVET-A--26/012--SE

# An Ergonomic Evaluation of Tilt Functionality in Reach Trucks

## Measuring Muscle Activation and Perceived Driver Experience

Sofia Davidsson

Supervisor: Birgitta Thorslund
Examiner: Peter Berggren

LINKÖPINGS UNIVERSITET

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# Abstract

In Europe, there are around six million individuals working as forklift drivers. These working environments typically include high stress on neck muscles, which can lead to pain and stiffness. Despite this, research pertaining to musculoskeletal issues in forklift drivers is limited, specifically in relation to how repeated and extreme neck extensions necessitated by their work affects them. The RT Reflex BT launched by Toyota Material Handling Europe, is a reach truck model developed with a tilting cabin aiming to decrease stress related to neck extensions and non-ergonomic working positions. Previous studies indicate that the tilting cabin can decrease neck extension by 20%, and that drivers show a positive attitude towards it. The present study aimed to deepen this understanding through a mixed methods approach, using both surface electromyography (sEMG) and semi-structured interviews analyzed thematically. The study recruited 22 participants used to working with the Toyota RT Reflex BT with tilt. The results indicate that using the tilting cabin may significantly decrease muscle activation when working at heights of eight and eleven-meters. The results motivate the tilting cabin as an ergonomically beneficial solution and further studies into how forklift drivers are affected by their work using sEMG.

# Acknowledgements

I would like to thank my advisor Brigitta Thorslund for supporting me throughout this project, and Christer Ahlström who acted as a second advisor and got me through my data analysis.

I would also like to thank Toyota Material Handling Europe for entrusting me with the project and giving me the opportunity to develop and shape it throughout. I would especially like to thank my advisor from Toyota Material Handling Europe, Alexandra Hjälmdahl, who has supported me throughout the entire duration of the project. Additionally, I would like to extend thanks to Susanne Wallgren and Lars-Eric Olsson who helped me with getting access to a truck, recruited participants and generally helped with the logistics of the data collection.

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# 1. Introduction

Forklift drivers make up a significant part of the workforce with six million forklift drivers in Europe alone (Flodin et al., 2018). They work mainly in warehouse and storage spaces – working in the logistics supporting companies worldwide in almost all fields. However, the way the drivers work can seriously affect their physical health, it has been reported that they can end up suffering from musculoskeletal pain (Gopanna et al., 2014; Ghafouri et al., 2024; Jimenez et al., 2023). In Sweden it has been reported that 20% of forklift drivers experience neck pain severe enough to limit them (Ghafouri et al., 2024). It has also been reported that they get into awkward positions in their working day (Viruet et al., 2008; Flodin et al., 2018), possibly resulting in serious musculoskeletal issues down the road. Despite this there is a serious lack of research in the musculoskeletal health of forklift drivers – particularly how their neck is affected by their work (Flodin et al., 2018).

There are solutions introduced which can assist in minimizing how often and to which degree drivers need to complete extreme head movements. For example, a camera can be mounted on the forks, allowing drivers to look at the camera monitor instead of extending their neck to see the pallet directly. However, there is a lot of trust needed for drivers to rely fully or primarily on the camera.

Another solution is the tilt cabin introduced by Toyota Material Handling Europe. Their BT Reflex reach truck features a tilting cabin, allowing drivers to look upwards without craning their neck as much. At least that is the idea, but there has not been ample research on how well this works in practice. Cederqvist (2000) has looked at the difference in degrees of head tilt occurs with and without using the tilt setting, finding that tilt resulted in a 20% decrease in degrees of extension. However, since then there has only been qualitative research.

Túren & Thorslund (2026) looked at forklift drivers' attitude towards using the tilt setting, finding it was positive in general and lessened strain and improved sightline. The tilting cabin has the potential to be a powerful tool, but its effectiveness needs to be confirmed and understood. Especially as the research is lacking in general when it comes to forklift drivers and their neck issues potentially stemming from their work, understanding how their neck is objectively affected is essential.

The quantitative method used by Cederqvist (2000) may have impacted the drivers' work as it was quite large. Furthermore, it looks at angles and not on the effect the movements have on the human body. To better understand the direct effect on the body another objective method was chosen: surface electromyography (sEMG). Ahlström et al. (2019) has looked at bus drivers and their muscle activation when evaluating a new assistive steering part in buses. With inspiration from his study sEMG was chosen as a reasonable quantitative method for this study.

In relation to the lacking research on how the neck extension effects reach truck drivers and potentially cause neck pain, conducting a study which looks directly at muscle activation is in the interest of all forklift drivers. Combining it with a qualitative method also substantiates previous research and puts the qualitative data in relation to quantitative data which contextualizes both datasets and brings nuance to the findings.

Thus, this study aims to combine surface electromyography (sEMG) collected with electrodes placed on the neck and shoulders of reach truck drivers with a semi-structured interview aimed

at understanding how they perceive the tilt cabin functionality. The goal is to better understand if and how the BT Reflex reach truck affects drivers, with a focus on their neck and subjective experience.

# 2. Research Questions

To reach the goal of the study three research questions have been formulated:

RQ1: Does the use of a tilt-cabin function in a reach truck result in less muscle activation in the sternocleidomastoid, neck extensor, and upper trapezius muscles of the operator?

RQ2: Is there a difference in mean muscle activation between eight-meter work and eleven-meter work with or without using a tilted cabin in reach truck work?

RQ3: Do forklift drivers' self-reported perception of the effect of using a tilt cabin reach truck align with their muscle activation while using one?

Three hypotheses are posited connected to each of the research questions; they are as follows:

H1: It is expected that the extension of the neck will not need to be as extreme with the tilted cabin as without, supported by Cederqvist (2000).

H2: It is expected that the mean difference in muscle activation between working at an eleven-meter height and working at an eight-meter height is the same with and without using tilt.

H3: Based on Túren (2026) drivers are expected to report a difference in how ergonomical the tilting truck feels and following that they perceive it to be helpful to issues or potential issues. This is expected to correspond to lessened muscle activation.

# 3. Background

The area of study is in a far-reaching, yet quite specific, field. Thus, understanding what constitutes as a forklift (3.1), how they are operated (3.2) and the specifics of the model being evaluated (3.3) is essential to understanding the context of the study.

## 3.1 Forklifts

**Figure 1**

*Examples of Forklifts*



(a) (b) (c)

*Note: Figure illustrating three types of forklifts; (a): a pallet jack, (b): a low-lifter with a platform, and (c): a reach truck. Images were taken from the Toyota Material Handling Sweden website: Våra truckar | Toyota Material Handling Sweden (2026-05-12).*

The term forklift includes everything from pallet jacks to highly complex reach trucks and pallet trucks where handling them requires a specific license (in Sweden). If we start at the perhaps least complex forklift, the pallet jack (Figure 1(a)), it is not driven but rather steered by an operator. They can carry between 2 to 3 tons, a considerable amount but not more than the operator can move by pulling the jack. It also has a simple construction – using a hydraulic pump which is activated by the operator moving the handle up and down to be able to lift cargo off the ground to height of about 20 cm (maximum).

If we move to more complex systems, we have forklifts which are driven by a standing operator such as low lifters (Figure 1(b)). In the model shown the operator stands on the raised platform and drives the truck, and they can decide in which direction they want to stand while driving. At this point a license is needed to operate the forklift in Sweden.

A reach truck is a type of forklift which is made for narrower spaces, and which can move laterally to the seated position of the operator (Babapour et al., 2012) (Figure 1(c)). Their main use is, as with all forklifts, to move cargo, often pallets. As such, they mostly appear in warehouse or storage yard environments – where they are of use. Unlike smaller forklifts which may be used when handling cargo on other vehicles, the reach truck is designed to work at greater heights. They also require a license in Sweden, with additional classification compared to what is needed for the low lifter.

When working with forklifts such as the reach truck or other seated forklifts there are different systems which companies can employ. Previously manual instructions had to be given to forklift drivers, now there are different warehouse management systems (WMS) available on the market which can automatically keep track of and delegate tasks to warehouse workers. Instead

of getting the instructions from a manager's mouth, drivers can instead see them on a screen and can sometimes see their queue of upcoming tasks as well.

## 3.2 Ergonomics of operating a forklift

Driving a forklift differs greatly from driving vehicles such as cars or trucks, as it can turn much more freely, and can thus be driven backwards with much more precision. Because of the innate design of a reach truck, if it is stacked high when depositing cargo at great heights the visibility is limited, which prompts drivers to extend and rotate their neck to better see what they are doing. It is more common for forklift operators to get into unnatural positions during their workday than office workers (Flodin et al., 2018).

There are helpful tools which can be implemented, such as cameras and lasers, to prevent or minimize the need for forklift operators to get into awkward or straining positions. Yet, forklift drivers still suffer from different types of musculoskeletal disorders (Gopanna et al., 2014; Ghafouri et al., 2024; Jimenez et al., 2023) and some links have been made to the ergonomical design of the forklifts. The most common areas for pain reported by forklift operators are in the lower back and neck and shoulders (Lu et al., 2018; Gopanna et al., 2014; Rolander et al., 2022; Jimenez et al., 2023). Gopanna et al. (2014) reported that 9 out of 47 of forklift drivers asked reported that the discomfort in their neck is significant enough to significantly affect their work, and 15 reported that the discomfort had a slight effect. This is supported by Ghafouri et al (2024) who reported that 20% of forklift drivers asked experienced limiting pain in their neck. Furthermore, the pain experienced by forklift drivers in their neck is reported to get worse after working (Ghafouri et al., 2024; Rolander et al., 2022). This implies that the work can be an aggravating if not causal factor.

Hoy et al. (2005) links lower back pain directly to twisting or bending the trunk forward and/or extending the neck significantly backwards. These are pointed out as the most prominent factors. Viruet et al. (2008) also links lower back pain in forklift operators to "awkward positions", this sentiment is echoed by other articles as either awkward positions or unnatural positions (Flodin et al., 2018; Túren & Thorslund, 2026).

To address these issues Toyota Material Handling Europe introduced the BT Reflex reach truck featuring a tilting cabin.

## 3.3 Toyota BT Reflex Reach Truck with Tilting Cabin

The BT Reflex reach truck with tilting cabin is a high-lifting truck which comes at different max capacities of 1400kg, 1600kg, 1800 kg and 2000kg (Toyota, n.d.). The first two can reach up to eleven-meters and the latter two can reach up to 13 meters and all feature a tilting cabin, see Figure 2. This function can be turned on or off from inside the cabin. Additionally, the truck can be fitted with different types of roofs for the visibility needed, a camera mounted to see the forks when working and a laser.

**Figure 2**

*The BT Reflex Reach Truck with Tilting Cabin*



*Note: The Toyota BT Reflex reach truck with Tilitng Cabin with the cabin in a tilted position (left) and in the non-tilted position (right). Taken from the Toyota Material Handling Sweden website: BT Reflex skjutstativ 2.0 ton med tiltande hytt - Skjutstativtruckar | Toyota Material Handling Sweden (2026-05-12).*

Previous work on the Toyota BT Reflex reach truck indicates that using the tilt function decreases head extension by 20% (Cederqvist, 2000). This previous work relied on a Nickometer, which is an ergonomic tool which can be used to measure head extenison and neck rotation. This tool is quite large and is connected to as band going aorund the participants head. Furthermore Cederqvist looked at subjective repsonses from participants by giving them a 10-point discomfort scale which he largely attributes to Borg (1980). The heights studies were 2m, 4m and 6m. From this he found that both the quantitative and qualitative data showed that the tilt cabin had a positive effect.

Túren & Thorslund (2026) also conducted a qualitative interview study using the TAM questionnaire related to the BT Reflex reach truck. The participants in the study were all used to the Toyota BT Reflex reach truck and the results showed that forklift operators were reported to that they perceive the tilt-cabin as ergonomically beneficial. They also perceived that their sightline was improved when using the tilt. It also ranked higher in perceived usefulness and perceived ease of use for participants. They were also open to increasing the tilt – with the reasoning that it would lessen pain.

# 4. Theory

The study relied on surface electromyography and the necessary theory to understand the method and results of the study, as well as previous work, is provided below.

## 4.1 Surface Electromyography

Surface electromyography (sEMG) is a summation of action potentials propagating along muscle fibers, which is the level of activation in a muscle, using an electrode located on the skin (Clancy et al, 2023). This electrode usually consists of silver (Ag), silver chloride (AgCl), silver/silver chloride (Ag/AgCl) or gold, electrodes made up of Ag or AgCl (or both) are preferred over gold as they are almost non-polarizable and thus have the smallest effect on the electrode-skin impedance. The electrode-skin impedance is essential for clear signals and is also affected by how the skin is prepared, so proper cleaning and shaving is essential. The size of the electrode and the composition of the gel used also affect impedance. The innervation zone of a muscle unit (MU), this is the area where single fiber action potentials travel towards both tendons when a healthy muscle gets activated.

When recording sEMG crosstalk can also be an issue – which is when activation of nearby muscles gets picked up by the electrode reading of the target muscle (Konrad, 2005), thus creating misleading results. This usually does not exceed 10-15%, however it can be an issue when measuring the trunk and shoulder muscles. If crosstalk is recorded it can be easily distinguished, and algorithms exist to ensure it is suitably removed from the final data being analyzed.

Bipolar sEMG is a version of sEMG where two electrodes are used for every muscle being measured, the output signal which is recorded is the difference between electric potential difference between the two electrodes (Stegeman & Hermens, 2007). The electrodes' dimension, orientation shape, position and distance to the other electrode all influence the output signal. However, since it takes two measurements it gives a better understanding of the activation in the target muscle and it is easier to filter out noise and crosstalk.

Data collected from electrodes needs to be normalized and one way to do this is by using a recording of a maximal voluntary contraction (MVC) by the participant (Burden, 2010). An MVC is when the participant completes the target movement to the furthest extent of their abilities, and it works well with healthy patients without muscular problems, and it can be either an isometric or dynamic movement. There is some discussion about the usefulness of an MVC and it is unclear when a dynamic MVC is preferable, however if the movement used represents a 100% activation of the muscles targeted it should be a good measurements to compare against. Because the MVC is used as a comparison of the real measurements taken later in the process, in order to understand how high the activation in real measurement is. It also helps with extreme outliers and in identifying misrepresentative data or malfunctions from the electrodes.

### 4.1.1 Previous Work in sEMG

Though no research was found where forklift operators range of neck extension has been evaluated using sEMG, there is previous work on overhead work where neck extension is measured. Neck extension refers to moving the head backwards to where the front of the neck is extended and overhead work is when someone has to perform a task above their own head. The movement of the neck in overhead work somewhat mimics that of truck drivers when working at great heights, however there are of course important differences. For one overhead works implicates that hands are going to be used which can affect the results of the studies, and

the goal of creating a headrest means they are not hindered. Truck drivers already have neck rest and are more restrained in how they can move as they are seated. Still, the extension of the neck is similar though less extreme and accompanied by hands.

Garosi et al. (2022) and Yao et al. (2024) both used sEMG to measure neck extension with the goal of building a neck support for overhead work. Garosi et al. (2022) looked at the trapezius and found a significant difference in muscle activity when their headrest was used, when using Ag/AgCl bipolar electrodes. Yao et al. (2024) looked at the sternocleidomastoids and they also found a significant difference with their headrest when using sEMG.

General studies on the movement of neck muscles using sEMG have also been completed (Swaminathan et al., 2016; Figas et al., 2023; Falla et al., 2002; Sommerich et al., 2000; Prisco et al., 2024). The type of electrodes used vary as do the muscles, but all the muscles focused on in this study has been included in the previous research; the neck extenders (Swaminathan et al., 2016; Prisco et al., 2024) the sternocleidomastoid (Falla et al., 2002; Figas et al., 2023; Sommerich et al., 2000) and trapezius (Figas et al., 2023; Sommerich et al., 2000). Supporting that sEMG is a reasonable method for studying these muscles.

Ahlström et al. (2019) also used sEMG to look at muscle activation in bus drivers. Ahlström et al. were able to discern differences in muscle activation depending on the equipment used for steering assistance in buses. Though the movement and muscles in focus were different, the trapezius was included but the other muscles were the deltoids, biceps and triceps, the driving aspect lends credibility to sEMG as a method for examining driving related muscle activation. Supporting that sEMG is also a reasonable method for driver situations.

## 4.2 Neck Musculoskeletal Disorders and Ergonomics

Musculoskeletal disorders is a term which includes over 150 diseases and conditions, which affect the locomotor system and are often characterized by pain (World Health Organization, 2022). The neck musculoskeletal disorders which cause neck pain include: muscle strain, ligament strain, myofascial pain, fibromyalgia, facet joint pain, internal disc disruption, somatic disfynction, spinal fracture, vertebral osteomyelitis, polymyalgia rheutmatica, atlantoaxial instability and atlanto-occipital joint pain (Meleger & Krivickas, 2007).

Ergonomical recommendations for reducing neck pain have been put forward by different authorities, both Swedish Agency for Work Environment Expertise (2025) and Occupational Safety and Health Administration (n.d.) recommend working with the neck in a neutral position. However, this neutral head position is not supported by all research (Wolff et al., 2024).

What has been show in research is that a flexion (forward bend) or extension (backwards bend) reduces negatively affects the neck (Ariëns et al., 2001; Ojukwu et al., 2020; Seghers et al., 2003; Van Det et al., 2009). Research on musculoskeletal pain and neck movement is somewhat lacking, focusing mostly on the effect of working at computers, using phones or in the surgical field.

## 4.3 Mental Workload

Mental workload has been defined as the attentional demands placed on an operator when performing a task and it is shaped by both exogenous and endogenous factors (Vidulich & Zang, 2015). Exogenous factors can be seen as external to the operator, for example system constraints, whereas endogenous factors are inherent to the operator, such as skill and ability.

The mental load of a task can therefore be identical for two operators but if one of them has a less developed skillset it may impose a larger mental load. Similarly tasks with different complexity may impose the same mental workload on different operators because of their inherent differences.

Though mental workload will not be measured explicitly in this study it is an intrinsic part of all work executed by a person. It will always be a factor affecting and affected by the work and changes in it, including the task performed in this study and thus it is contextually relevant.

# 5. Method

The recruitment of participants and data collection of the study occurred during several months, and the specifics of how it was carried out and who was included are reported below.

## 5.1 Participants

The study recruited 22 participants ranged between 21 and 56 years, where 13,64% identified as female while 81,82% identified as male and 4,54% did not report their gender.

On average the participants reported driving a forklift or reach truck for 31,77 hours every week (min: 0, max: 40) and they reported having had their forklift certification for an average of 13,34 years (min: 3, max: 40). One participant reported that they drove a forklift less often than every month, one reported driving every other week and the rest reported driving every day. All the participants had previously driven a reach truck with a tilt cabin, and they all reported using it in their current work.

## 5.2 Recruitment

The study used convenient sampling where participants were recruited through Toyota Material Handling Europe's connections – both in their factory in Mjölby and through contact with their customers. Presentations of the general purpose (developing the tilt reach truck) and an explanation of how electrodes would be applied were given to ensure participants did not go in completely blind.

Ten out of 22 participants were recruited from Toyota Material Handling Europe's own production, and 12 were from other companies. Participants were given a few gifts from Toyota Material Handling as a thank you for participating – though this was not advertised when recruiting – and they were treated to lunch or coffee after they were done with the study. They were also visiting during work hours and did not lose any income from participating nor did they have to use their free time.

## 5.2 Study Design

The study was a within group study conducted at the Toyota Material Handling Europe facilities in Mjölby. The truck used was a BT Reflex reach truck provided by Toyota Material Handling Europe. The focus of the study was to understand the muscle activation in the neck of and experience for the drivers when using the tilt function of a reach truck – as well as the relation between those measurements. The condition being changed was if the participant was driving with tilt or without. The muscle activation was measured using sEMG and the driver perception through a semi structured interview. The study is in line with ethical practices put forward by Swedish Research Council (2024).

Before the method was finalized a pilot was conducted which ended up affecting the final method. This was primarily in relation to the task given, which changed from moving a pallet between heights to moving it between the ground and different heights and a turning moment was discarded. It was also clear that tape was needed to secure the electrodes better to participants.

## 5.3 Equipment

The study utilized electrodes (5.3.1), the specific truck used (5.3.2) and recording equipment used during the drive and during the interview. The specifics are presented below.

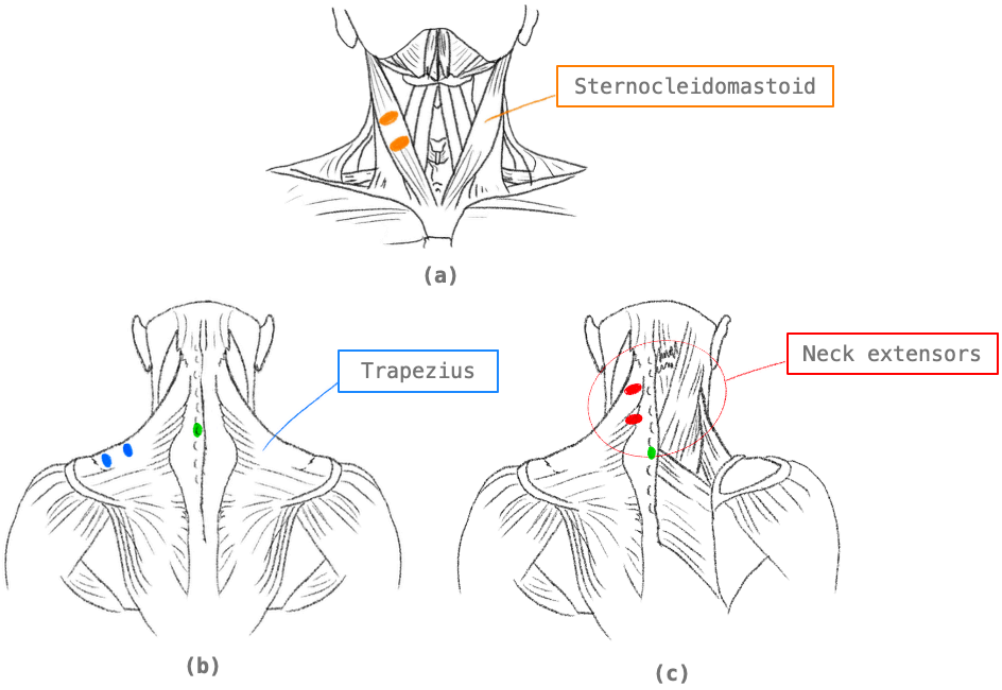
### 5.3.1 Electrodes

The electrodes used were the Neuroline 720 Ag/AgCl electrodes from Ambu. The electrodes are self-adhesive, but an elastic weaved tape was used to further secure the electrodes during the test. For each participant a total of 13 electrodes were used and were connected to a Vitaport 3 amplifier by TEMEC. Before the electrodes were applied the skin was shaved and then cleaned with the abrasive Nuprep Skin Prep Gel made by Weaver and Company.

Two electrodes were placed on six individual muscles of three muscle types, and with one centimeter spacing between their active sites, see Figure 3. The placement was based on SENIAM (2026) and Konrad (2005) and supported by previous research (Swaminathan et al, 2016; Prisco et al., 2024; Falla et al., 2002; Figas et al., 2023; Sommerich et al., 2000).

**Figure 3**

*Placement of Electrodes*




*Note: An illustration of which muscles were targeted and where electrodes were placed- The top illustration (a) show the sternocleidomastoid and the two electrodes can be seen in orange. The bottom left (b) illustration shows the trapezius and the electrode placement can be seen in blue. The bottom right (c) shows the neck extensors and the electrode placement can be seen in red. Both (b) and (c) include the neutral electrode in green. Traced from Britannica, and altered: https://www.britannica.com/science/human-muscle-system.*

For the upper trapezius muscle SENIAM (2026) was followed both for general placement and procedure in placing the electrodes. The distance between the C7 vertebrae and the acromion was measured and the half-way point was marked, the electrodes were then placed with their

active site 1 cm apart on one side each of the mark. They were placed along the muscle, see Figure 3(a).

The placement of electrodes on the sternocleidomastoid muscle was based on Konrad (2005), and the procedure of placing the electrodes was based on the anatomy of the sternocleidomastoid muscle (Cleveland clinic, 2023) and general recommendations from SENIAM (2026). As well as placement from previous studies (Figas et al., 2023; Yao et al., 2024). The distance between the sternal end of the clavicle to the mastoid was measured and the halfway point was marked. The electrodes were then placed along the measured line, one with its active site 1cm away from the mark upwards along the line and one 1cm away downwards on the line, see Figure 3(b).

Finally for the neck extensors the placement was based on Konrad (2005) and the procedure was based on Nobe et al. (2022). The C7 vertebrae was located and the location of C4 was felt or approximated based on this and a mark was made on top of it. From there 2cm were measured and marked on each side of the original mark and one electrode was placed above and one below this mark, both with their active area 1 cm away from the mark and along the muscle. See Figure 3(c).

The neutral electrode was placed on the protrusion of the C7 vertebrae, as recommended by SENIAM (2026) (seen in Figure 3(b) & 3(c)).

### 5.3.2 Reach Truck

The truck used was a Toyota BT Reflex reach truck with tilting cabin with a max capacity of 1400 kg and a max reach of eleven-meters in height. The truck had a transparent roof, chosen based on what was available at the time of the study. It was fitted with a camera on the forks and a laser. The monitor for the camera was located toward the top of the inside of the cabin, more on the right side. A camera was also mounted in the top left corner of the cabin.

Each participant was asked to adjust the seating to their preference. The reach truck did not have an arm rest on the side of the seat closest to the entry point.

When using the tilt setting the cabin started tilting when the forks reached two meters in height.

### 5.3.3 Recording Equipment

To record video data and sound three pieces of technology were used during the data collection: a GoPro, a laptop and a pair of eye-tracking glasses. The GoPro was of the model GoPro Hero10 Black. It was mounted using a magnetic mount to the roof of the truck. It had a wide-angle feature which made it easy to adjust the framing to include the participants pressing the button.

For the audio recording during the thematic interviews a laptop was used, this laptop was provided by Toyota Material Handling Europe, and the audio recordings were only ever stored on this computer. The microphone was the one built into the laptop. The sound quality was tested before recording started and it was clear enough to record the interviews.

The eye-tracking glasses were the Tobii Pro Glasses 3 (Tobii, n.d.), the collected material was not used in this study, but they were used by participants when collecting the data for this study.

## 5.4 Procedure

The participants were asked to sign a non-disclosure agreement when relevant, and a consent form for the study (see Appendix 1). Each participant was assigned a participant ID which all their collected data was marked with after signing to keep them anonymous. They were then equipped with electrodes and were introduced to what was going to happen in the study, and then moved to the lab area. Once in the lab area the electrodes were tested and the participants were shown the route they were going to drive and informed about how they were meant to press a button available to them, and that one of the test leaders was going to be instructing them throughout the drive. This instruction followed a predetermined schedule, an excerpt can be seen in Table 1 and the full table can be found in Appendix 2.

**Table 1**

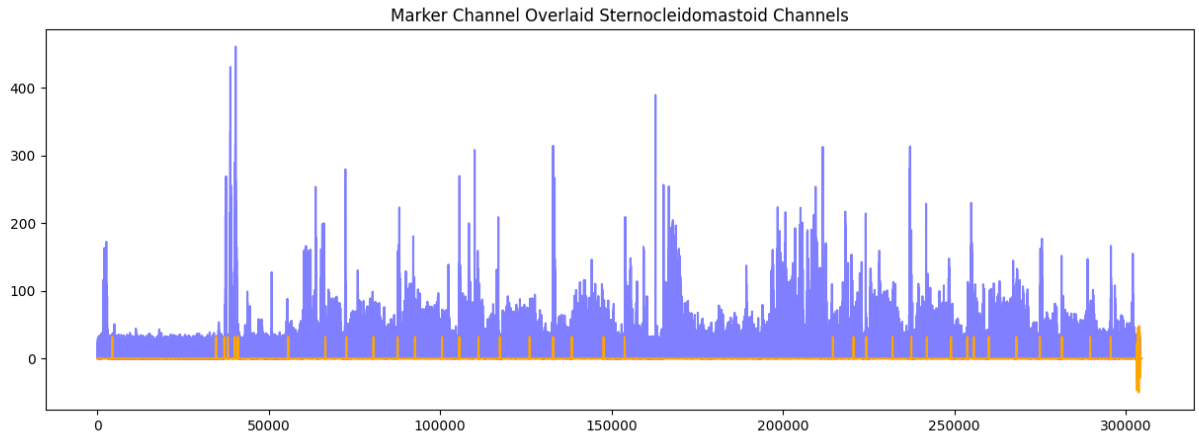
*Schedule for Pressing the Button*

| Press of button (sequentially numbered) | Action marked |
|---|---|
| 13 | After pallet is put back 8m |
| 14 | After forks are at floor level |
| 15 | Before lift |
| 16 | Before pallet is lifted 11m |

The button corresponds to its own channel in the amplifier and each press of the button created a mark in that channel, see Figure 4, which corresponded to the schedule and marked the beginning of a segment.

**Figure 4**

*Illustration of Marker Signal*




*Note: Figure showing the marker channel in orange, each press of the button corresponds to a raised mark in the channel (seen as vertical stripes), overlaid on top of one of the channels for the sternocleidomastoid muscle (blue). The final orange mark shows the button being disconnected from the amplifier.*

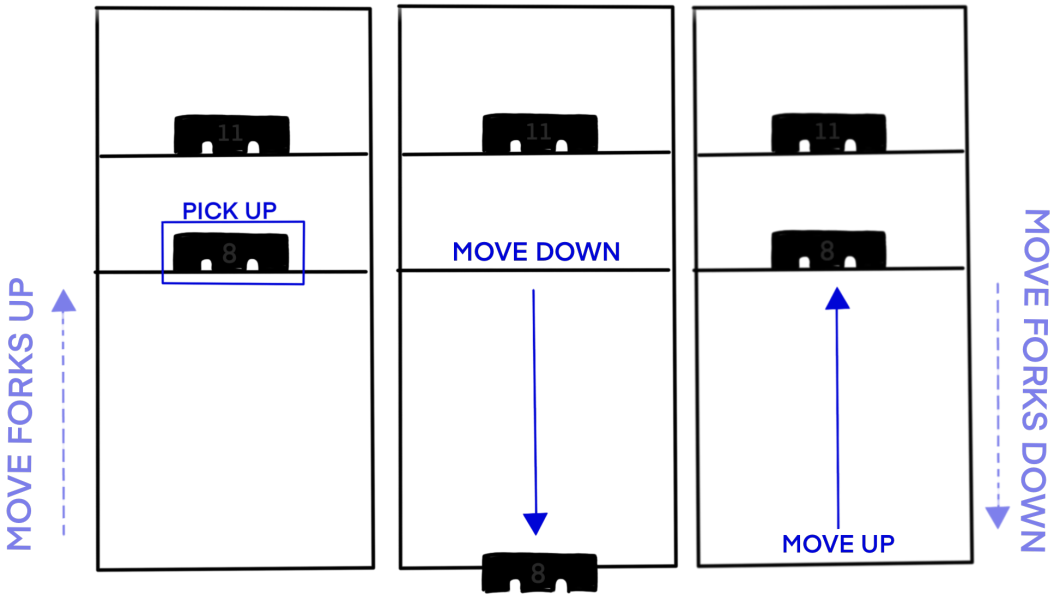
The participants were then asked to get into the forklift and adjust the seat to their preference and the camera in the forklift was turned on. At this point they were also supplied with a pair of eye-tracking glasses if they were not wearing regular glasses, used for the separate study conducting data collection simultaneously as this study. The eye-tracking data was not collected or analyzed in this study.

They were then asked to relax as much as possible for two minutes, then to tilt their head upwards as much as possible and finally to contract their muscles as much as possible. These two movements were done to collect a reading for a maximum voluntary contraction (MVC), the data used in the end was from participants contracting their muscles as much as possible. Finally, they were given the button to the electrodes and told to press it before they started driving, and to press the button and start driving when they were ready. They then performed their task, half of them started with the tilt setting turned on and half with it turned off, changing in the middle. When this change occurred a NASA-TLX survey was also administered for most participants for the data collection in a separate study – thus the resulting data from this was not analyzed in this study either.

The participants' task consisted of picking up, taking down and then placing a pallet back up where they picked it up several times. Participants were asked to first pick up a pallet on a shelf at 8m height, and then take it down to the ground, before putting it back up on the same shelf. Then they were asked to take the fork all the way to the ground again, before repeating the task with a pallet on a shelf at eleven-meters. Then they were asked to return to the starting point, where the test leader helped the participants change tilt setting, before they repeated the same tasks again. Figure 5 shows the steps of the task for the pallet at an eight-meter height, which were then repeated for the eleven-meter height.

**Figure 5**

*Task Illustration*




*Note: Illustration of the task. The figure goes from right to left – beginning with forks being moved up, then the pallet gets picked up, then moved all the way down, moved back up to its original position and the forks were then lowered (illustrated to the left-most side of the figure).*

The task was done in an aisle where two pallets were used to delimit the area where the driver could drive, a lane which measured 3,5 meters across, this was designed to be similar to what a forklift driver may encounter in their day-to-day work based on on-site expertise. The setup was put up in a warehouse environment and the two pallets weighed 500 kg each.

Throughout the task they were also asked to use the button supplied to them in the truck which was connected to the sEMG amplifier and marked the data. This was done according to the predetermined schedule introduced previously, see Table 1 and Appendix 2, and the test leader informed the participant of every time they had to press the button. As the truck was open and the environment fairly quiet, this was simply done by the test leader talking loudly to the driver while standing behind the reach truck. The video material collected by the GoPro was also adjusted to capture the participant pressing the button.

The test leader did not give any other instructions except for when the participant should press the button and answering questions if the participant posed them. Questions answered were related to the next step task or if they should press the button. If they asked if they should press the button they were told no and informed again that the test leader would inform them every time, they should press the button. The test leader did not comment on driving style and participants were free to drive as they wished as they completed the task in order ot capture their authentic use of the reach truck.

After the driving portion of the study was completed, the participants were let out of the truck and the recording of sEMG signals was stopped and the electrodes were disconnected from the amplifier. Participants using eye-tracking glasses were also asked to take those off. The same NASA-TLX survey was then administered – still for a separate study – and a semi structured interview was held with the participant, the questions for this can be found in Appendix 3.

Once the survey and interview were completed participants were helped to remove electrodes and the study was finished.

## 5.5 Data Analysis

The data analysis consisted of three parts, two for the sEMG data and one for the interview data. The sEMG data had to be processed (5.5.1) before it could be used to garner statistical results (5.5.2). The data from the interview portion of the study was analyzed through a thematic analysis (5.5.3).

### 5.5.1 Pre-Processing of sEMG Data

The data was processed through several steps. Firstly, the means of the data was removed, then a Butterworth bandpass filter was used to filter the data, the settings of the filter were based on Mello et al. (2007). The data was then whitened and rectified.

A Welch analysis was also conducted for all participants to find any outward noise from the truck at a certain Hz, the data was also inspected visually to find if any clear discrepancies existed and if they were extreme enough to warrant excluding the data. Including if any electrodes had become unattached or had other issues during the study – those recordings were then excluded.

As the participants were asked to perform a maximum voluntary contraction (MVC), in this case contracting their muscles as much as they could, the max activation recorded during this for each muscle was set as the benchmark for 100% activation. The rest of the data could then be related to this benchmark by dividing mean and max of each relevant segment by the MVC and then multiplying it with 100.

When analyzing relevant segments, the marker signals were used with an added buffer to ensure that most of the movement related to pressing the button was excluded. The marker signal was

a separate channel connected to a button and at the beginning of each segment the button was pressed, creating a mark in the channel, see Figure 4 in 5.4. The buffer was created by timing three times the button was pressed for each participant and getting the mean time. Half of the time was then added before the marker and half of it was added after.

Several participants had missing or additional marks from pressing the button when they had not pressed the button hard enough or when a disturbance which warranted an extra press of the button occurred. To rectify this the video from the GoPro was used to find where marks needed to be removed or added. To add markers the time of the missing marker was taken as well as the time of the previous and following press of the button, then the marker was added at the same distance from each of these values and was added to the collection of other marker values. Some participants also had marks at the end of their marker channel from the button being disconnected, see Figure 4, which were removed as well.

The results from the sEMG results were either used as they were after the pre-processing or after being normalized to the MVC max value recorded for each muscle. When data was normalized the neck extensors were not included as the MVC did not work, and when neck extensors were included three participants neck extensor data was excluded. Their data for other muscles were still included, but after a visual inspection the neck extensor data was found to be abnormal.

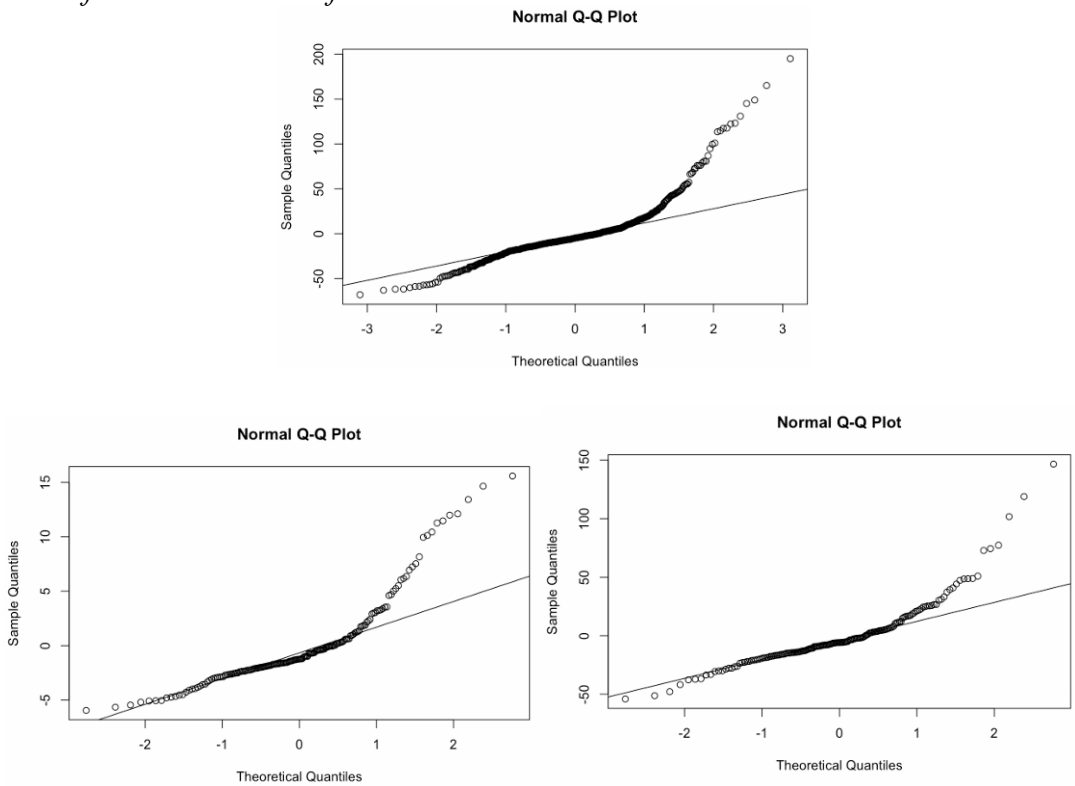
The data was processed using python and the full code can be found in this open repository: https://github.com/sofda00/tilt-truck-data-processing.

### 5.5.2 Data Characteristics

The Shapiro-Wilk normality test found that the data is not normally distributed when it has been normalized to a max-value ($W = 0{,}833$, $p = 3{,}73\ e^{-16}$), nor when it is not normalized ($W = 0{,}830$, $p = 2{,}60\ e^{-16}$). It was not normally distributed when the data from the neck extensors was removed and not normalized ($W = 0.84875$, $p = 3.204e^{-12}$) nor when the data was normalized ($W = 0.83415$, $p = 7.163e^{-13}$). Figure 6 shows the curve of the data visualized.

**Figure 6**

*QQ-Plots of the Distribution of Data*

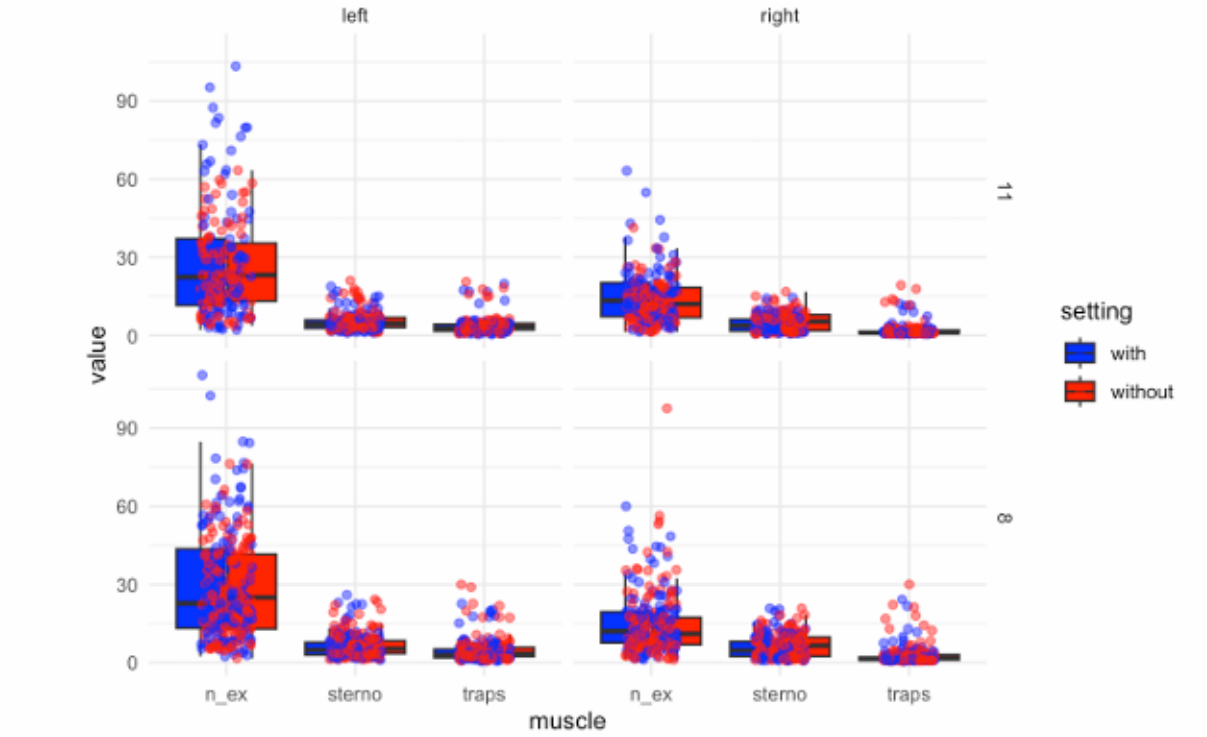



*Note: The top plot shows a QQ-plot of the data including all muscles and not normalized against the MVC max, the bottom left shows a Q-Q Plot of the data from the sternocleidomastoid and the trapezius when not normalized against the MVC max and the bottom right shows it when normalized.*

The data also had several outliers, see Figure 7, however as the group consisted of only 22 individuals no outliers were removed.

**Figure 7**

*Outlier Visualization*




*Note: Visualization of the outliers in the data. The settings are with tilt (blue) or without tilt (red) and showing the different working heights of eleven-meters (top) and eight-meters (bottom). The bottom notation stands for the muscle groups; n_ex stands for the neck extensors, sterno for the sternocleidomastoids and traps for the trapeziuses.*

The Welch analysis did not result in any clear disturbances, and no extra filter was applied.

### 5.5.3 Statistical Analysis

For the statistical analysis of the sEMG eight Linear Mixed Models (LMM) were conducted, comparing the results for different heights, tilt settings, muscles and participants. LMM's are longitudal statistical models, including within-subject error correlation, and they are a more robust option compared to an ANOVA or MANOVA (Arnau, 2012). Because of the outliers and the data not being normalized (see 5.5.2) a more robust option was necessary, and the LMM accounts for the participants effect on the results by including it as a random effect.

The analysis was performed between each participant's results when driving with and without the tilt and segments were separated by the schedule for when the button should be pressed (See Table 1 and Appendix 2) to identify different movements in/aspects of the task and compare the results. Participant was used as a random factor in all models and tilt setting (with/without) was always a factor, where driving without tilt was compared to driving with tilt. Either all three muscles groups were included or the data was normalized and the neck extensor muscles excluded. In each model one level of each factor is set as comparative, it is against this level which the other levels of the factors and the other factors are compared. Table 2 shows a summary of the setup of LMM's.

**Table 2**

*Summary of the Setup of Each LMM*

| LMM number | Muscle groups included | Normalized data | Additional factors | Comparative Factor |
|---|---|---|---|---|
| 1 | Neck extensors, sternocleidomastoid, trapezius | No | None | Neck extensors |
| 2 | Sternocleidomastoid, trapezius | Yes | None | Trapezius |
| 3 | Neck extensors, sternocleidomastoid, trapezius | No | Muscle side (left, right) | Neck extensors, Left |
| 4 | Sternocleidomastoid, trapezius | Yes | Muscle side (left, right) | Trapezius, Left |
| 5 | Neck extensors, sternocleidomastoid, trapezius | No | Task (up, down, work) | Neck extensors, Up |
| 6 | Sternocleidomastoid, trapezius | Yes | Task (up, down, work) | Trapezius Up |
| 7 | Neck extensors, sternocleidomastoid, trapezius | No | Initial Setting (started with tilt, started without tilt) | Neck extensors, Started with |
| 8 | Sternocleidomastoid, trapezius | Yes | Initial Setting (started with tilt, started without tilt) | Trapezius, Started with |

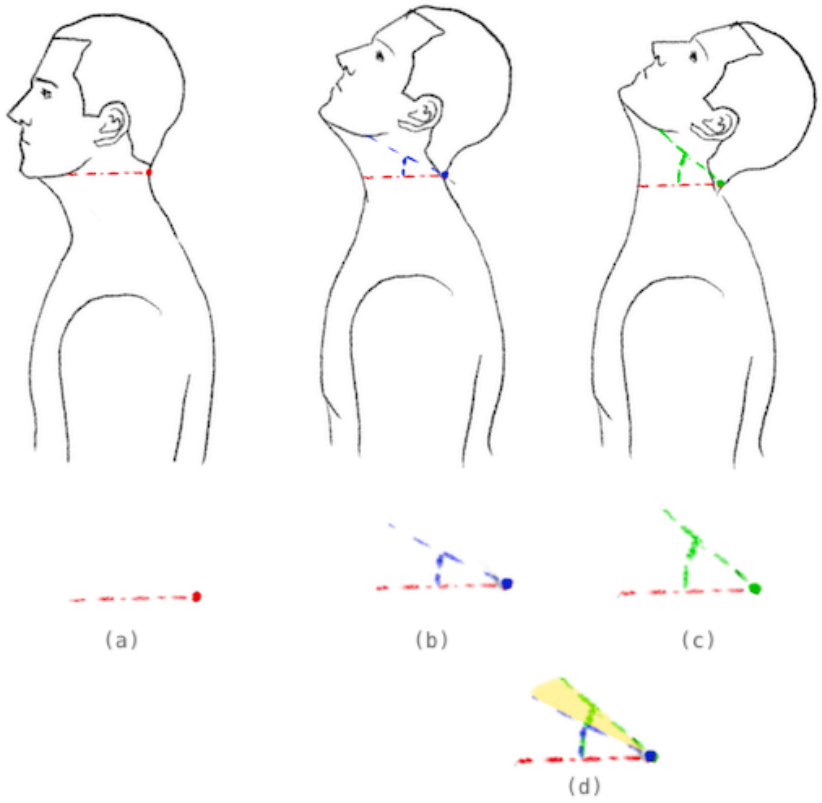
Before conducting the LMM's a Shapiro-Wilk normality test on residuals was conducted. However, the results did not exclude the use of the LMM as a method but were used to contextualize the results.

The max value in each segment was also analyzed using bar plots. The analysis was restricted to 95% of the data, to exclude extreme outliers. The values were restricted to 95% of the highest data points and the difference in max-values for with tilt or without tilt was compared. Type of movement, initial setting or sides were not included in this analysis.

A comparison of the mean difference between the eleven-meter work and the eight-meter work was also completed. The area of interest was the mean difference between eight-meter work versus the eleven-meter work with tilt and the difference in activation between the eight-meter work and the eleven-meter work without the tilt setting. Figure 8 illustrates the difference of interest, this difference was taken from each participant and the mean was used for analysis. The mean activation from both sides were used. Two comparisons were done; one with all muscles using the mean activation and one with the sternocleidomastoid and trapezius using the mean activation normalized against the MVC max. These comparisons will be referred to as the mean difference between heights henceforth.

**Figure 8**

*Difference Between Heights Illustration*




*Note: Figure illustrating the difference in neck movement between eight-meter height and eleven-meter height. The illustration is not based on a measured difference in degree of neck extension, but is for illustration purposes only. (a) shows a neutral position, and a line indicating the degrees accompanying the neutral position, (b) shows the neck extended to look at an eight-meter height, with the change in degrees illustrated alongside the initial line for the neutral position. (c) shows the extension to look at the eleven-meter height, with the accompanying change in degrees illustrated alongside the line for the initial neutral position. (d) illustrates the difference between the levels of extension (highlighted in yellow) alongside the line for the neutral position.*

All statistical tests were done using R in RStudio and the full code is in an open repository (https://github.com/sofda00/tilt-truck-data-processing).

### 5.5.4 Thematic Analysis

A thematic analysis following Alvinius et al. (2023) was done on the qualitative data collected through a semi structured interview held after the driving portion of the study. The interview questions can be found in Appendix 3, but they were not the only ones asked nor were they asked if the topic was already discussed. The analysis was inductive – where themes were identified as they appeared, and no specific questions or themes were brought into the analysis to the best of the analyzer's ability.

The interview audio was recorded and manually transcribed. As the interviews were semi structured all questions were not asked in the same order and topics varied somewhat between participants. Because of this the topic found in the interviews, from questions asked or unprompted, were ordered and the data was organized against which topic it related to. All the data was then put into a spreadsheet, and the order was randomized to get rid of lingering bias before themes were identified.

The individual interview transcriptions were then analyzed one by one – noting the important themes. These notations were then compiled together to get a bigger picture of the themes prevalent in the full span of interviews.

# 6. Results

The results are divided by sEMG results (sections 6.1-6.3) and thematic analysis results (6.4). The statistical results from the LMM's conducted are presented first, followed by max values and height comparisons to further shed light on the trends in muscle activation. Finally the results from the thematic analysis is presented. It should be noted that only 21 out of 22 participants are included in the thematic analysis as one did not partake in an interview.

## 6.1 Linear Mixed Models

Several Linear Mixed Models (LMM's) were conducted with all muscles, however as a result of the MVC max not working with the neck extensors, LMM's using data normalized against the MVC max did not include the neck extensor muscles. Using normalized data is standard, thus it is used for the trapezius and sternocleidomastoid data, but comparing data which has not been normalized against normalized data would severely impact the results. Therefore, all three muscle groups are compared without being normalized, to be able to get any results for the neck extensors.

Furthermore, three participants neck extensor data was excluded as the electrodes malfunctioned.

### 6.1.1 Muscles and Setting

A linear mixed-effects model (LMM) was fitted using the mean activation per segment, with tilt setting and muscle type as fixed effects and participant as a random effect. This model identified two significant fixed effects ($p < 0.05$). This LMM aimed to focus on the tilt setting and showed that the sternocleidomastoid and trapezius muscles both exhibited lower activation than the neck extensors (sternocleidomastoid: ES = -25.581, $p <$ 2e-16; trapezius: ES = -10.137, $p =$ 1.69e-05). The interaction between no-tilt and the trapeziuses approached significance ($0.05 < p < 0.1$), indicating a tendency for higher activation in the trapeziuses when tilt was not used (ES = 6.452, $p = 0.0505$).

When using data normalized against the MVC max and excluding the neck extensors, a second LMM revealed two significant effects ($p < 0.05$). Specifically, not using tilt resulted in higher overall muscle activation compared to using tilt (ES = 0.64814, $p = 0.000253$), and the trapezius showed lower activation than the sternocleidomastoid (ES = -2.72459, $p <$ 2e-16).

Table 3 shows the significant and almost significant effects (full table in Appendix 4).

**Table 3**

*Significant and Marginally Significant Effect of First LMM*

| Factors | Estimate Std. | Error | df | *t*-value | *p*-value |
|---|---|---|---|---|---|
| | | Not Normalized | | | |
| | | REML: 30385.1 | | | |
| (Intercept) | 52.300 | 4.074 | 29.451 | 12.837 | 1.36e-13 *** |
| Sterno | -25.581 | 2.353 | 3001.514 | -10.874 | < 2e-16 *** |
| Traps | -10.137 | 2.353 | 3001.514 | -4.309 | 1.69e-05 *** |
| Without:Traps | 6.452 | 3.298 | 2997.073 | 1.956 | 0.0505 . |
| | | Normalized | | | |
| | | REML: 10560.9 | | | |
| (Intercept) | 5.91691 | 0.75715 | 21.88592 | 7.815 | 9.01e-08 *** |
| Without | 0.64814 | 0.17682 | 2087.00000 | 3.666 | 0.000253 *** |
| Traps | -2.72459 | 0.17682 | 2087.00000 | -15.409 | < 2e-16 *** |

*Note: Sterno = Sternocleidomastoid, Traps = upper trapezius, Right = right side of body, Without = Without tilt setting. Codes: 0 < *** < 0,001 < ** < 0,01 < * < 0,05 < . < 0.1. Full table in Appendix 4.*

### 6.1.2 Side

An LMM using the mean activation per segment and including tilt setting, muscle type, and muscle side as fixed effects and participant as a random effect, identified six significant fixed effects ($p < 0.05$) and one marginally significant effect ($0.05 < p < 0.10$). The significant main effects indicated that the right side of the body exhibited lower activation compared to the left (ES = -61.3556, $p$ < 2e-16). Furthermore, both the sternocleidomastoids and the trapeziuses exhibited lower activation compared to the neck extensors (sternocleidomastoid: ES = -53.5918, $p$ < 2e-16; trapezius: ES = -24.6607, $p$ < 2e-16). Several significant interactions were also observed: the combination of no tilt and trapezius activation resulted in higher activation (ES = 9.1613, $p$ = 0.0215), as did the interaction between right side and the sternocleidomastoids (ES = 56.0380, $p$ < 2e-16), and between the right side and the trapeziuses (ES = 29.0640, $p$ = 3.8e-13). The setting was approaching significant, indicating that no-tilt lead to a lower muscle activation overall (ES = -5.0699, $p$ = 0.0824).

When data was normalized against the MVC max, and neck extensors excluded, a second LMM revealed two significant effects ($p < 0.05$). Specifically, the trapeziuses showed lower activation (ES = -1.8123, $p$ = 8.25e-14), as did the interaction between the trapeziuses and right side (ES = -1.8246, $p$ = 9.66e-08). One effect was marginally significant ($0.05 < p < 0.1$), indicating higher muscle activation when tilt was not used (ES = 0.4523, $p$ = 0.0608).

Table 4 shows a summary of significant and almost significant effects for both LLM's including sides as a factor (full result table in Appendix 5).

**Table 4**

*Significant and Marginally Significant Effect of Second LMM*

| Factors | Estimate Std. | Error | df | *t*-value | *p*-value |
|---|---|---|---|---|---|
| | | Not Normalized | | | |
| | | REML: 29410.4 | | | |
| (Intercept) | 82.9697 | 4.2452 | 34.7489 | 19.544 | < 2e-16 *** |
| Without | -5.0699 | 2.9184 | 2991.0533 | -1.737 | 0.0824 . |
| Right | -61.3556 | 2.9184 | 2991.0533 | -21.024 | < 2e-16 *** |
| Sterno | -53.5918 | 2.8296 | 2992.7587 | -18.940 | < 2e-16 *** |
| Traps | -24.6607 | 2.8296 | 2992.7587 | -8.715 | < 2e-16 *** |
| Without:Traps | 9.1613 | 3.9840 | 2991.0533 | 2.299 | 0.0215 * |
| Right:Sterno | 56.0380 | 3.9840 | 2991.0533 | 14.066 | < 2e-16 *** |
| Right:Traps | 29.0640 | 3.9840 | 2991.0533 | 7.295 | 3.8e-13 *** |
| | | Normalized | | | |
| | | REML: 10408.3 | | | |
| (Intercept) | 6.0727 | 0.7661 | 22.9443 | 7.926 | 5.11e-08 *** |
| Without | 0.4523 | 0.2411 | 2083.0000 | 1.876 | 0.0608 . |
| Traps | -1.8123 | 0.2411 | 2083.0000 | -21.024 | 8.25e-14 *** |
| Right:Traps | -1.8246 | 0.3409 | 2083.0000 | -18.940 | 9.66e-08 *** |

*Note: Sterno = Sternocleidomastoid, Traps = upper trapezius, Right = right side of body, Without = Without tilt setting. Codes: 0 < *** < 0,001 < ** < 0,01 < * < 0,05 < . < 0.1Full table in Appendix 5.*

### 6.1.3 Task

An LMM based on mean activation per segment, with setting, muscle type, and task as fixed effects and participant as a random effect, identified three significant effects ($p < 0.05$) and one marginally significant effect ($0.05 < p < 0.10$). The significant main effects showed lower activation for the sternocleidomastoids (ES = -26.1982, $p$ = 1.00e-10), the trapeziuses (ES = -8.9838, $p = 0.0261$), and the downward movement (ES = -10.2243, $p = 0.0143$). The interaction between the sternocleidomastoid and the downward movement approached significance and indicated higher activation in this condition (ES = 10.0353, $p = 0.0780$).

When using data normalized against the MVC max, and excluding the neck extensors, a second LMM revealed two significant effects ($p < 0.05$) and one marginally significant effect ($0.05 < p < 0.1$). Not using tilt resulted in higher activation compared to using tilt (ES = 0.75724, $p$ = 0.0133). The interaction between no-tilt and the down movement was marginally significant and indicated in lower activation (ES = -0.80760, $p = 0.0620$). The trapezius muscle remained a significant factor associated with lower activation (ES = -2.66599, $p <$ 2e-16).

Table 5 shows a summary of significant and almost significant effects for both LLM's including task as a factor (full result table in Appendix 6).

**Table 5**

*Significant and Marginally Significant Effect of Third LMM*

| Factors | Estimate | Std. Error | df | *t*-value | *p*-value |
|---|---|---|---|---|---|
| Not Normalized | | | | | |
| REML : 30298 | | | | | |
| (Intercept) | 53.4865 | 4.7303 | 53.4003 | 11.307 | 8.67e-16 *** |
| Sterno | -26.1982 | 4.0369 | 2986.6111 | -6.490 | 1.00e-10 *** |
| Traps | -8.9838 | 4.0369 | 2986.6111 | -2.225 | 0.0261 * |
| Down | -10.2243 | 4.1699 | 2985.0721 | -2.452 | 0.0143 * |
| Sterno:Down | 10.0353 | 5.6925 | 2985.0721 | 1.763 | 0.0780 . |
| Normalized | | | | | |
| REML: 10551.9 | | | | | |
| (Intercept) | 5.98558 | 0.77744 | 24.32610 | 7.699 | 5.64e-08 *** |
| Without | 0.75724 | 0.30578 | 2079.00000 | 2.476 | 0.0133 * |
| Traps | -2.66599 | 0.30578 | 2079.00000 | -8.719 | < 2e-16 *** |
| Without:Down | -0.80760 | 0.43244 | 2079.00000 | -1.868 | 0.0620 . |

*Note:. Sterno = Sternocleidomastoid, Traps = upper trapezius, Right = right side of body, Without = Without tilt setting, Work = Work with pallet at height, Down = Moving pallet down. Codes: 0 < *** < 0,001 < ** < 0,01 < * < 0,05 < . < 0.1. Full table in Appendix 6.*

### 6.1.4 Initial Setting

An LMM based on the mean activation per segment, with setting, muscle type, and initial setting as fixed effects and participant as a random effect, identified six significant fixed effects ($p < 0.05$) and two marginally significant effects ($0.05 < p < 0.10$). Three out of the significant effects were associated with lower muscle activation: not using tilt compared to using tilt (ES = -7.535, $p = 0.03157$), the sternocleidomastoids (ES = -33.354, $p < 2e\text{-}16$), the trapeziuses (ES = -16.613, $p = 6.55e\text{-}07$). The remaining three significant effects were associated with higher muscle activation and involved interactions: no tilt x trapezius (ES = 10.313, $p = 0.02613$), starting without tilt x sternocleidomastoid (ES = -16.613, $p = 6.55e\text{-}07$) and starting without tilt x trapeziuses (ES = 11.917, $p = 0.01139$). Among the marginally significant effects, starting without tilt indicated lower muscle activation (ES = -15.250, $p = 0.07614$), whereas the interactions no tilt x sternocleidomastoid (ES = 8.358, $p = 0.07140$) and no tilt x starting without tilt (ES = 8.949, $p = 0.06396$) indicated higher muscle activation.

When the data were normalized against the MVC max, and the neck extensors were excluded, a second LMM identified two significant effects ($p < 0.05$). The trapeziuses were associated with lower muscle activation (ES = -2.5853, $p < 2e\text{-}16$), while the interaction between not using tilt and starting without tilt was associated with higher muscle activation (ES = 0.8149, $p = 0.0216$).

Table 6 shows a summary of significant and almost significant effects for both LLM's using initial setting as a factor (full result table in Appendix 7).

**Table 6**

*Significant and Marginally Significant Effect of Fourth LMM*

| Factors | Estimate Std. | Error | df | *t*-value | *p*-value |
|---|---|---|---|---|---|
| Not Normalized | | | | | |
| REML: 30338.4 | | | | | |
| (Intercept) | 60.291 | 5.672 | 29.549 | 10.629 | 1.30e-11 *** |
| Without | -7.535 | 3.503 | 2992.069 | -2.151 | 0.03157 * |
| Sterno | -33.354 | 3.333 | 2999.854 | -10.009 | < 2e-16 *** |
| Traps | -16.613 | 3.333 | 2999.854 | -4.985 | 6.55e-07 *** |
| Start Without | -15.250 | 8.277 | 27.696 | -1.842 | 0.07614 . |
| Without:Sterno | 8.358 | 4.634 | 2992.069 | 1.804 | 0.07140 . |
| Without:Traps | 10.313 | 4.634 | 2992.069 | 2.225 | 0.02613 * |
| Without:Start Without | 8.949 | 4.829 | 2992.069 | 1.853 | 0.06396 . |
| Sterno:Start Without | 14.770 | 4.706 | 2996.208 | 3.138 | 0.00172 ** |
| Traps:Start Without | 11.917 | 4.706 | 2996.208 | 2.532 | 0.01139 * |
| Normalized | | | | | |
| REML: 10547.8 | | | | | |
| (Intercept) | 6.0868 | 1.0491 | 29.549 | 5.802 | 9.62e-06 *** |
| Traps | -2.5853 | 0.2389 | 2084.0000 | -10.821 | < 2e-16 *** |
| Without:Start Without | 0.8149 | 0.3544 | 2084.0000 | 2.300 | 0.0216 * |

*Note: Sterno = Sternocleidomastoid, Traps = upper trapezius, Without = Without tilt setting, Start Without = Started without the tilt setting. Codes: 0 < *** < 0,001 < ** < 0,01 < * < 0,05 < . < 0.1. Full table in Appendix 7.*

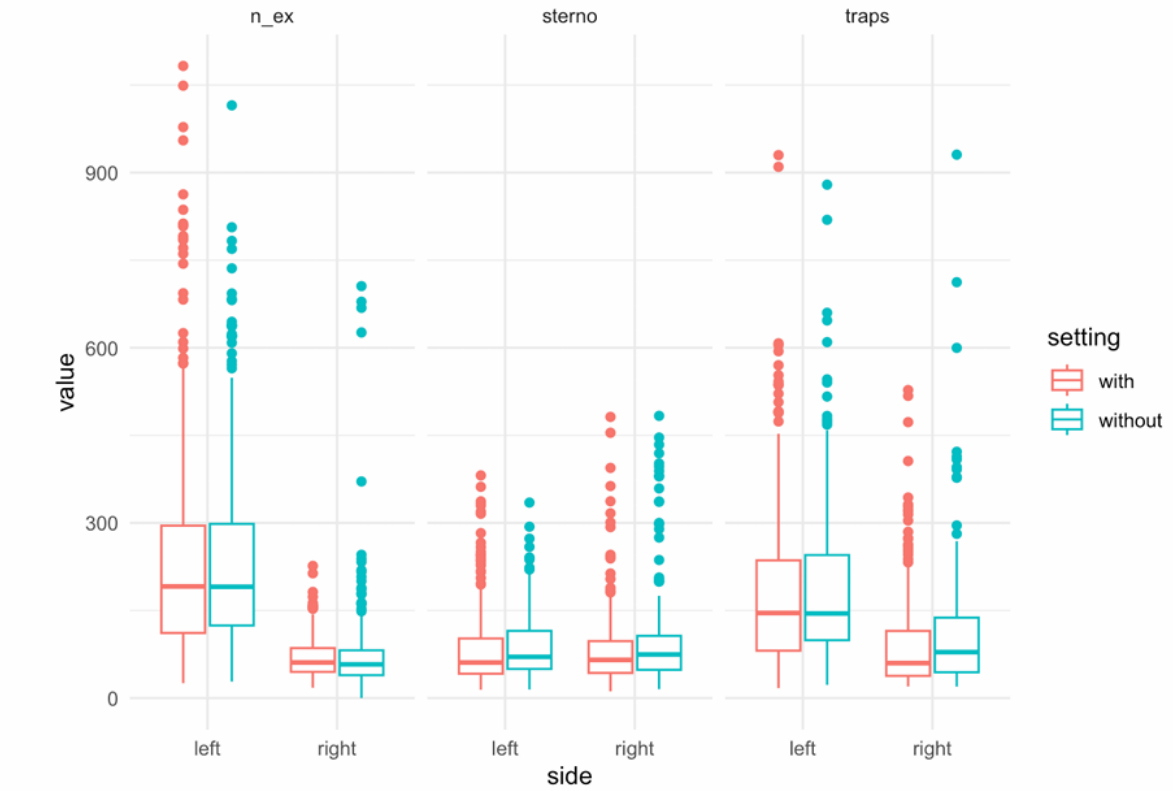
## 6.2 Max Values

A max value was computed for each segment, muscle and side and were normalized, both the initial value and the normalized one were then plotted, see Figure 9 and Figure 10. The max value was not the absolute max but limited to 95% of the signal to exclude extreme outliers. When tilt was used, activation was lower overall, with slightly higher activation on the right side.

In contrast, for both the sternocleidomastoid and the trapezius muscles, the tilt setting resulted in lower activation. The sternocleidomastoid exhibited similar activation levels between sides, with a slight predominance on the right side, whereas the trapezius showed clearly higher activation on the left side. See Figure 9.

**Figure 9**

*Max Values, Not Normalized*



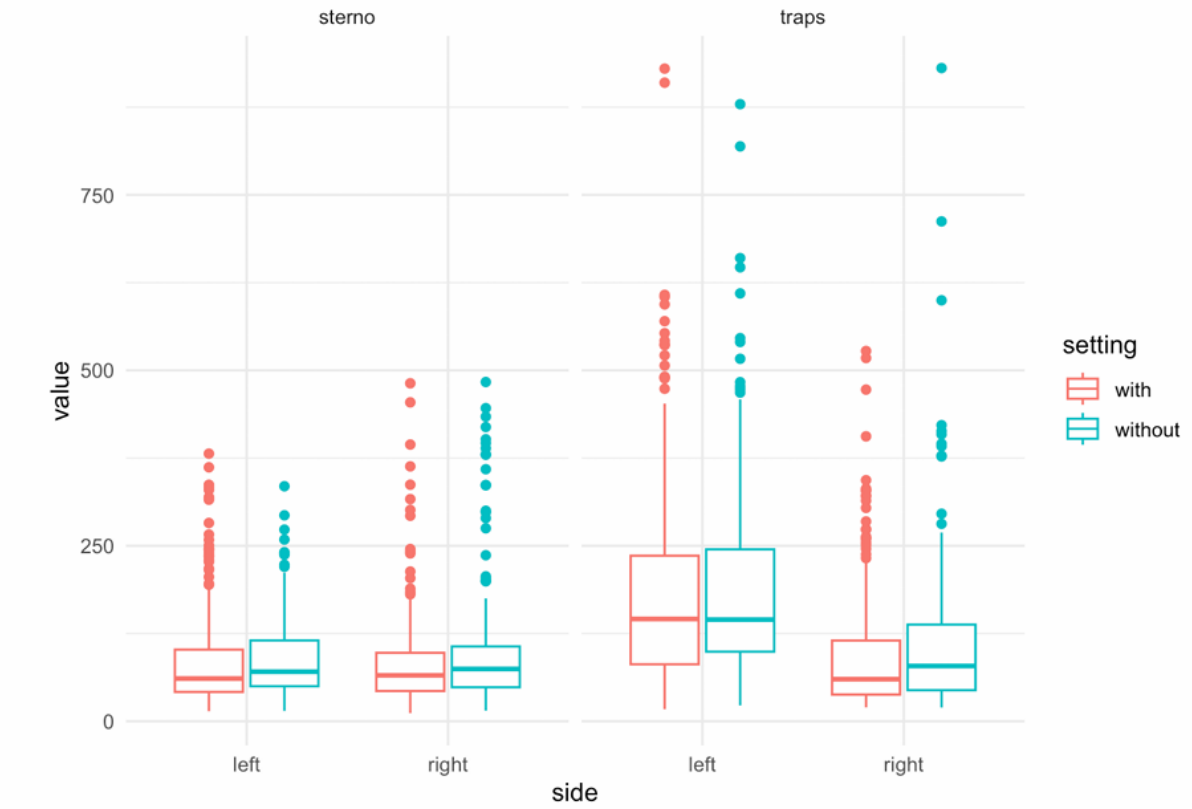

*Note: Max value of the 95% percentile of data, not normalized, for all three muscles: neck extensors (right), sternocleidomastoid (middle) and trapezius (left). Red visualizes values when tilt was used, and blue when tilt was not used. Each muscle group has also been divided into the left and right muscle.*

For the normalized data, the neck extensor muscles showed higher activation with tilt on the right side, but without tilt on the left side, and overall activation was higher on the left side.

For both the sternocleidomastoid and trapezius muscles, the tilt setting resulted in slightly lower activation. The sternocleidomastoid showed very similar activation between sides, with a small predominance on the right side, whereas the trapezius exhibited clearly higher activation on the left side. See Figure 10.

**Figure 10**

*Max Values, Normalized*




*Note: Max value of the 95% percentile of data, normalized against the muscle max, including the sternocleidomastoids (left) and the trapeziuses (right). Red visualizes values when tilt was used, and blue when tilt was not used. Each muscle group has also been divided into the left and right muscle.*

## 6.3 Mean Difference Between Heights

The difference in muscle activation between work at eight meter and eleven-meter height was calculated by subtracting activation with tilt from the corresponding activation without tilt for each movement. Movements were categorized as up, up (second), work, work (second), down, and down (second). The full results can be found in Appendix 8.

Using the mean difference between height for each movement, the results showed that the effect of tilt differed across muscles. In 17 of the 36 total movement conditions, the use of tilt resulted in a larger difference in activation between the two heights. This pattern occurred for at least one movement type in each muscle group. The height-related differences in activation are illustrated in Figure 11.

The mean difference between heights when using data of the mean activation during each movement normalized against the MVC max, it is shown to be different when tilt was used and when it was not, for both the sternocleidomastoids and the trapeziuses. It shows no clear pattern between movements, but 12 out of the total 24 movements showed that using tilt resulted in a bigger difference in activation when working at different heights. This phenomenon occurred for some movement in each muscle. These differences are plotted below (Figure 12).

The mean differences between heights were illustrated using bar plots, with results presented separately for tilt and no-tilt conditions (Figures 11 and 12). For each movement, the mean activation at eight-meters was subtracted from the mean activation at eleven meters. Consequently, negative values indicate higher activation at eight-meters, while positive values indicate higher activation at eleven meters.

Across both non-normalized and MVC-normalized data, the plots do not reveal any clear or consistent pattern related to mean difference between heights and tilt usage. For several muscles and movements, activation was lower at eight-meters than at eleven-meters, whereas the opposite pattern was observed in other cases.

Similarly, the muscles and movements for which tilt resulted in a larger mean difference between heights showed no systematic structure. In the non-normalized data, some movements exhibited positive differences for both tilt and no-tilt conditions, others showed negative differences in both conditions, and several showed mixed effects depending on tilt usage. These patterns varied across muscles and movements, indicating that the influence of tilt on mean difference between heights was movement- and muscle-specific rather than uniform.

Overall, while tilt increased the mean difference between heights for a subset of movements, these effects were heterogeneous and did not follow a consistent directional trend.

**Figure 11**

*Overview of Activation Differences, Not Normalized*

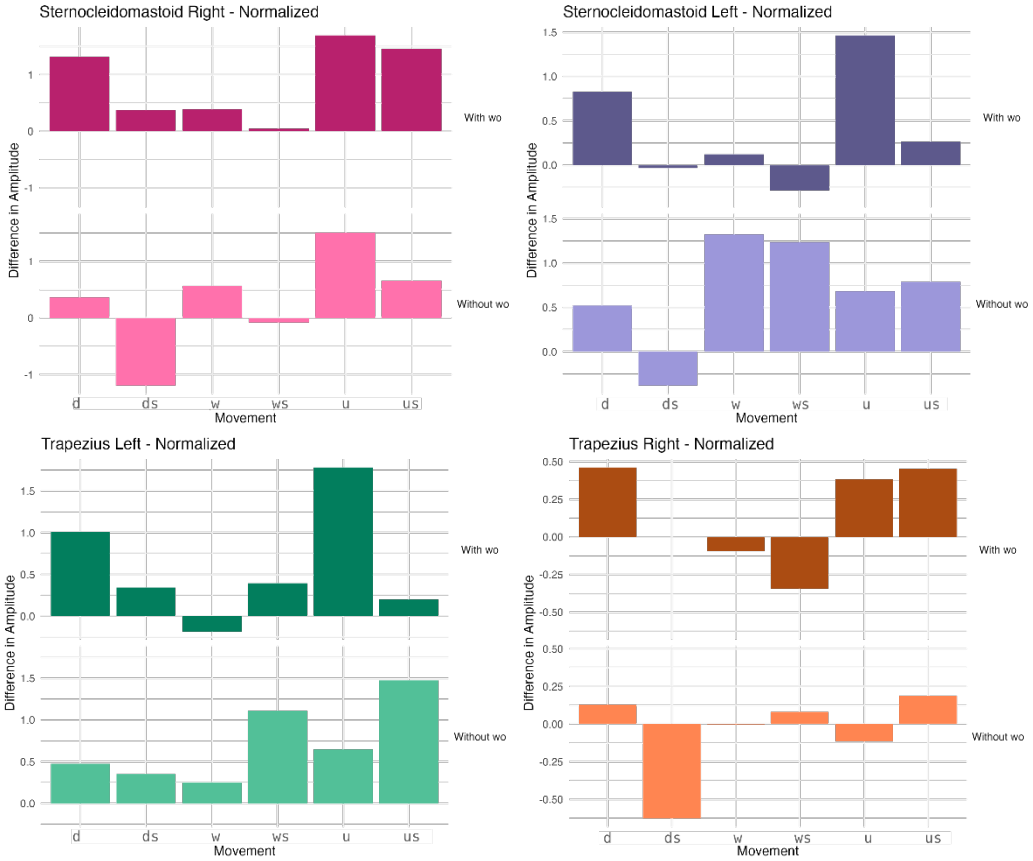



*Note: Barplots for the difference in activation for the mean activation of each movement of the Sternocleidomastoid left (top left), Sternocleidomastoid right (top right), Trapezius left (middle left), Trapezius right (middle right), Neck extensor left (bottom left) and Neck extensor right (bottom right).X-axis codes: d = down, ds = down second, w = work, ws = work second, u = up, us = up second..*

Of the 12 cases in which MVC-normalized data were used and tilt resulted in a larger height-related difference, six showed positive mean differences for both tilt and no-tilt conditions, indicating higher activation at eleven-meters (see Figure 12). One case showed negative mean differences in both conditions, indicating higher activation at 8 m. In two cases, the mean difference was negative when using tilt but positive without tilt, whereas three cases showed the opposite pattern. These mixed results further illustrate the absence of a consistent directional effect of tilt on height-related activation differences.

**Figure 12**

*Overview of Activation Differences, Normalized*




*Note: Barplots for the difference in activation for the mean activation of each movement of the Sternocleidomastoid left (top left), Sternocleidomastoid right (top right), Trapezius left (bottom left) and Trapezius right (bottom right), normalized against the MVC max. X-axis codes: d = down, ds = down second, w = work, ws = work second, u = up, us = up second..*

## 6.4 Thematic results

All thematic results are taken from transcriptions of interviews held in Swedish, for readability only excerpts of full quotes in translated to English is presented here, the original Swedish transcription and the English translation is available in full in Appendix 9. The interviews analyzed were on average 2:04 minutes long (min: 01:09 min, max: 03:40 min). 21 interviews were conducted – one participant was not interviewed because of technical issues during the study. All the interviews were conducted individually.

The first division of themes in the interview answers was largely dependent on which questions and themes had been brought up by the interviewer and which had been answered to a larger degree. The categories found were: Attitude towards tilt, bodily pain in their day-to-day, follow-up on bodily pain, attitude when prompted about bodily pain, negative perceptions of tilt, preference for/regular use of tilt and last additions.

From each of these the relevant parts of the transcripts were added to a table, with one table per category and analyzed for relevant themes. From this several themes were discovered

throughout, mainly the tilt function was appreciated, and it was described as better (P11), more pleasant (P13), and it was mentioned throughout by several participants that the tilt improved their sightline. This was specifically described as improving the view of the sides of the pallet. The effect on the neck muscles with and without tilt was also brought up a lot, mostly unprompted but also after being asked specifically.

Relevant themes were defined as ones with three or more mentions, keeping in mind that all questions were not explicitly asked to all the participants. The most prevalent themes within each category can be seen in the table below:

**Table 7**

*Overview of Identified Themes*

| Category | Prevalent themes | Number of mentions |
|---|---|---|
| Attitude towards tilt | Prefer tilt | 12 |
| | Feel higher discomfort in neck without tilt | 10 |
| | Better sightline with tilt | 7 |
| | Big difference | 5 |
| | Less strained with tilt | 4 |
| | Easier with tilt | 3 |
| Bodily pain in their day-to-day | Neck pain/discomfort | 5 |
| | No followed by remarking tilt is better for the neck | 3 |
| Follow-up on bodily pain | Neck pain/discomfort | 5 |
| | Back pain/discomfort | 3 |
| Attitude when prompted about bodily pain | Neck pain/discomfort | 6 |
| | shoulder pain/discomfort | 3 |
| Negative perceptions of tilt | Nothing | 8 |
| | Tight workspaces present challenges | 4 |
| Preference for/regular use of tilt | Preference for tilt | 14 |
| | More comfortable with tilt | 3 |
| Last additions | Enjoy tilt with their specific truck | 3 |

These themes were then analyzed further by going back to the original transcripts to contextualize and pick out all the instances when the themes were mentioned. From this the context of when and how the themes were brough up by the participant, how many times the themes were brought up and how many of the participants touched on different themes and if they were positive or negative. It is of note that for the last row in Table 7 that the other participants answered that they had nothing more to add – or added something which was not directly relevant, such as the temperature being too low in their warehouse.

As the interviews were semi-structured some of the participants touched on subjects without being asked, whereas some were not talkative and needed more questions. Thus, the themes

were somewhat categorized depending on if the answer was “prompted” by a question which may have influenced the participant, or if it was more of an unprompted theme.

This was especially prominent in relation to the effect of the tilt function on the neck, shoulders and back. Participant 18 had an unprompted answer which included how it felt in the neck when asked about their general experience of driving with or without tilt, note that the participant usually drives without tilt:

*I thought somehow it felt much better in one way to drive the one where you don’t tilt with it, I felt more in the neck and all of that. (P18)*

However, participant 9 did not mention anything body-related when asked about their general experience and was thus prompted by the interviewer.

*A little in neck and in shoulder but ah, it usually goes away as the week passes (I:mm) but course’ it’s always there but it is not something that’s straining so. But as I said I think you need to keep moving a little somehow to work at it otherwise it is probably hard a- then you can get stiff, I would say at least. (P9)*

Another prominent theme was the fact that the tilt function improved the driver’s sightline. This came up both in the beginning when asked about their general experience, but also throughout the interview in several cases. This improvement was described by two participants as improving the view of the sides of the pallet, see below, and others mentioned seeing better as well.

*Yeah of course a bit at the top it is better, leaned back you see the sides and edges a little better. (P6)*

*Big difference (I:mm) *unclear* leaning you can kind of see the edges. (P3)*

It was also mentioned that the improved sightline helped with putting the forks in the pallet explicitly, but one participant also described how the tilt setting helped them to avoid having to search for where the forks are meant to go in.

One participant (P12) also brought up how the tilt setting explicitly made the work feel safer:

*You get more ehhm of a handle ha no but it feels safer when you don’t have to sit so bent with the neck or something. (P12)*

All in all, 12 participants mentioned preferring the tilt setting when initially asked about their attitude towards the task they had completed. At the same point ten participants mentioned feeling more the movement more in their neck when the tilt setting was turned off and seven participants mentioned that their sightline was improved with the tilt setting. Five participants said the difference was big and four of them mentioned that the tilt setting made them more relaxed or less strained.

Several participants also mentioned that the tilt made things easier – this mentioned as the tilt making it easier to complete the task, as it making it easier to see parts of the pallet and as the work without the tilt being described as harder. It was mentioned by three participants in the towards the task, and it was also mentioned when asked about if they usually drive and prefer driving with tilt and when asked for additional thoughts.

Several participants also mentioned how pain in the body and neck occurs in the beginning of driving. This was talked about as if beginning of driving meant the beginning of a career or after a break in driving. One participant also credited the evolution of the tilt trucks to lessened pain.

To summarize the general attitude from the participants was a positive attitude towards using the tilt setting, mainly rooted in it being better for the body and improving their sightline. All participants seemed to be in agreeance on the benefit of the tilt setting, and none of them expressed a need or want for a change to a non-tilt reach truck in their work. The only reoccurring negative take was that it is harder to use in tighter aisles, but this was also attributed to badly built warehouses than an issue with the tilt truck by one participant. One participant did also comment on the tilt being unpleasant (P17) for drivers who are not used to it.

# 7. Discussion

The discussion will be separated into three sections; Result discussion, method discussion and future work. In the first section results are contextualized with previous research and against the research questions, which are: *Does the use of a tilt-cabin function in a reach truck result in less muscle activation in the sternocleidomastoid, neck extensor, and upper trapezius muscles of the operator?* (RQ1). *Is there a difference in mean muscle activation between eight-meter work and eleven-meter work with or without using a tilted cabin in reach truck work?* (RQ2), and *Do forklift drivers' self-reported perception of the effect of using a tilt cabin reach truck align with their muscle activation while using one?* (RQ3).

The method discussion then further contextualizes this by highlighting parts of the process which can cast doubt upon results, or which otherwise needs to be discussed. Leading finally to future work and in which direction the research could continue based on results and methodological insights.

## 7.1 Result Discussion

The results indicated that using tilt resulted in lower activation, and it was reported to feel better and improve the drivers' sightline. There is some nuance though most of the results agree. In almost all linear mixed model (LMM) results where setting was significant, using tilt resulted in lower muscle activation. These results were prominent when using normalized data and only including the muscle and when including the task, and an interaction with the trapezius muscles when including sides. This interaction was almost significant when sides were not included, still showing that tilt resulted in lower activation and when sides were included in the LMM, both with and without normalized data, an almost significant effect was found.

The only time that using tilt was not shown to result in lower activation, but it was still significant, was when using non-normalized data and including initial setting. However, the same LMM results show that the interaction of not using tilt and the sternocleidomastoids, the trapezius muscles and the initial setting resulted in higher activation, significant for the interaction with the trapezius and almost for the others. Indicating that the effect of the setting is somewhat ambivalent in these results. The effects may also be that the results of the neck extensors are skewing the results, as they are in general more extreme than the other muscles. Furthermore, the right neck extensor muscle showed a slightly higher activation for the with tilt setting when looking at the max values of each muscle, which no other muscle shows. Indicating that they stand out from the other muscles, and when conducting the same LMM using normalized data (and excluding the neck extensors) only the interaction with the initial setting significant, resulting in higher activation. This further supports that the effect may be tied to the neck extensors, and it may be skewing the results. It does pose the question though if the neck extensors are particularly affected by the initial setting, as the significance is still there.

Though the tilt is not shown to result in lower activation in all LMM's conducted, the results support the first hypothesis to some extent– where it is posited that without tilt results in higher muscle activation. This hypothesis is also supported by the max-value bar plots, the trapezius and sternocleidomastoid muscles had consistently higher values when not using tilt, though not for the right neck extensor as mentioned previously.

This could be related to the results from the thematic analysis, where an improved sightline was brought up by several participants. It is possible that the improved sightline led to participants looking more and using their neck extensors because they were able to see more of the pallet.

One participant did also mention the tilt cabin making the work safer – which could be connected. Being able to see the pallet better may very well lead to looking more closely and thus being more meticulous when working. It could also be related to the quality of the data recorded from the neck – but the reliability of using sEMG for the neck extensors should be studied before any judgement is made on the data quality.

It is also interesting to discuss whether the mental load was increased as the sightline was reportedly improved. It is possible that the possibility of looking at more of the pallet made operators feel as though they had to look more meticulously – thus increasing their mental load – and possibly resulting in using their neck muscles more. To understand this the results from the study collecting eye-tracking and NASA-TLX data in the same data collection sessions as this study could be used to further contextualize the findings.

Beyond the tilt setting the results from the LMM's showed that the right side had a significant effect, resulting in lower activation, but its interaction between right and the sternocleidomastoids and right and the trapeziuses resulted in higher activation. When using normalized data only the interaction between the right side and the trapezius had a significant effect, resulting in lower activation. As the normalized data does not include the neck extensors it is possible that the right side of the neck extensors are a lot less activated than the left neck extensor. This is supported by the max values results, where it is apparent that the left side of the neck extensors have a much higher activation for both settings.

This could be a direct effect of the tilt truck not having an arm rest, which can be fitted into the reach truck but was not for this study. As the right side of the operator is the side where the controls are, with an accompanying arm rest, this support may be the explanatory factor for the difference. This would need to be explored further before judgements are made. Another cause may be the fact that the monitor for the camera and the mirror is fitted inside of the cabin. To see past them it may be easier to look outside of the cabin on the left side where the cabin is entirely open. Again, this would have to be looked at closely before judgement calls are made.

The task was also somewhat significant when not using normalized data, where the down movement was significant and resulted in lower activation, when compared to the up movement. However, down also had an almost significant interaction with the sternocleidomastoid muscles where the interaction resulted in higher activation. The interaction between the sternocleidomastoid and the up movement is hard to place, but the significance of the down movement could be related to the monotony of the task. When the pallet was moved down it was going to land on the ground, whereas when it was being moved up it was in preparation for either picking up or putting back the pallet. It may be that the drivers had to look more closely or keep their eyes more on the pallet and shelf when moving up, in preparation for picking up or returning the pallet, which is a much more precise task than moving to the floor.

The interaction between down and not using tilt was also almost significant when using normalized data, resulting in lower activation. This is interesting as well, but it may be an effect of the tilt again supplying a better view, and the drivers looking more closely when the opportunity is given. It could also stem from the drivers feeling safer during the tilt, though only expressed outright by one participant during the interviews and thus looking at other things and straining their neck. Research on where the drivers are looking would have to be necessary to understand why this phenomenon occurs.

The initial setting also had somewhat of an effect beyond its interaction with not using tilt, where the interaction between starting without tilt and the muscle type was significant when not using normalized data. With both the sternocleidomastoid muscles and the trapezius muscles it had a significant effect which increased activation. However, the effect of just starting without tilt only had an almost significant, decreasing, effect on activation. It should be noted that participants were used to using tilt, and significantly higher activation may be a result from not being used to a reach truck without tilt. It may have led participants to feel like they had to compensate for a worse sightline by using more extreme movements. The almost significant effect does not support this, however as it is almost significant it does not carry the same weight, further research would have to be done to understand if this effect should be accounted for. It would be interesting to see if drivers who are not used to the tilt experience the same effects, and if a study with a larger population gets the same almost significant result.

The most consistent factor was the muscle group, where the sternocleidomastoid and trapezius muscles always resulted in lower activation compared to the neck extensors. Which is also in line with the max-value results, where the neck extensors have a notably higher max activation. This supports previous findings that forklift operators suffer from musculoskeletal pain (Gopanna et al., 2014; Ghafouri et al., 2024; Jimenez et al., 2023), as neck extension has been identified as being a cause of musculoskeletal pain (Hoy et al., 2005).

When looking at the difference in activation between heights 17/36 of the movements in the non-normalized data had a higher activation difference with tilt, whereas 12/24 had a higher activation when using normalized data. This does not indicate that the tilt setting had higher activation – but rather that the difference in movement was bigger. The bar plots do not show any pattern for when the activation difference is higher with tilt, nor do they show any pattern in the means of these instances in whether the eight- or eleven-meter mean is larger. Thus, this measurement does not tell us too much and does not support or oppose the second hypothesis, which expected no real difference, as the differences do not seem to be consistent. There was also no clear pattern in how big the difference was at all, which might mean that there isn't a clear one. Considering the relatively small sample size, the not-normalized data and the number of outliers, there may be too many outstanding factors to get a good and robust understanding of the situation.

Some other interesting results from the bar plots are the fact that the eight-meter height sometimes resulted in higher activation than the eleven-meter height. There was no clear pattern here either, but the phenomenon occurred throughout muscles and movements. Most prominently in the right neck extensor, both with and without tilt, which is very interesting. As the setting does not seem to be the deciding factor something about the right side of the neck seems to be more affected by working at and bringing down the forks from the eight-meter height. This might be a result of using cameras more at the greater height – or looking less because the sightline is inherently worse at eleven-meters than at eight, when looking with the naked eye. It is an interesting phenomenon which could be explored further.

For the third hypothesis, which posits that the participants will prefer the tilt setting, the thematic analysis supports it and the results are in line with the sEMG results which is also posited in the hypothesis. Though, it is of note that the thematic results are a lot more decisive and clearer in the favor of using the tilt than the sEMG results, which are more varied and the difference between using tilt and not was often small if prominent. This could plausibly be related to the fact that musculoskeletal disorders are not only caused by muscle strain (World Health Organization, 2022). There is a real possibility that the difference reported is not caused

by or only by a difference in muscle activation, but in difference in strain in other parts of the neck which are felt more keenly by operators.

It must also be said that the discrepancy may of course stem from operators wanting to give the "correct" answers to questions or feeling as though the interviewer had a view which they should uphold. In this case the thematic results would not accurately represent the operators experience, and the difference in results would be explained. However, the thematic results are also in line with previous research done by Túren & Thorslund (2026), where the tilt is preferred and sightline is a theme brought up just as it was in this study. Which does not support this reasoning but rather supports the robustness of the results. It is of note that both this study and Túren & Thorslund's were conducted with participants who were used to the tilt trucks in their daily work and at least one participant in each study brought up that the tilt may be intimidating or uncomfortable for novices.

The thematic analysis was also in line with previous work showing that forklift drivers reported more pain after driving (Ghafouri et al., 2024; Rolander et al., 2022). The thematic analysis shows that several participants comment on pain after a day's work, despite having access to tilt. Though, it should be noted that it was not reported how often they used the tilt, just that they used it in their current work. It may be seldomly used, or the reach truck may be used at heights where the tilt is minimal, or it may be programmed to start at a higher height than the one used in the study. But it could also indicate that despite the statistical results indicating lower muscle activation as an effect of using the tilt, to some extent, it can still mean that even with tilt musculoskeletal pain can become an issue. This could be because the tilt should be more extreme to avoid pain, as requested by participants in Túren & Thorslund (2026), or because the muscle activation is not the critical factor.

## 7.2 Method Discussion

The methods used in the study were anchored in previous work, but there were still some issues which occurred. The reasons for these issues varied, but their occurrence and impact need to be discussed in relation to the study and its results.

Starting with the equipment used, and specifically the placement of the two electrodes recording the Sternocleidomastoid. The placement was based on previous research, but two of them were placed to record the movement in the sternocleidomastoid. However, though these muscles are close to the surface there is the platysma muscle above it. The platysma muscle is a thin surface level muscle (Hoerter & Patel, 2023). Though the electrodes were measured to sit at the middle of the Sternocleidomastoid and was based on previous work studying the Sternocleidomastoid, the platysma covers the same area. It is therefore not entirely certain which of the muscles has been primarily recorded. The question is whether it is entirely relevant in this study, for future research on the sternocleidomastoid or platysma it is, but the aim of this study was to find out how and if the neck is affected by tilt and non-tilt settings on the BT Reflex reach truck. Whether the activation is primarily from one frontal muscle or another is not detrimental to the interpretation of the results, the most important factor is that electrodes have been placed consequently, and that activation of the neck is being recorded.

The neck extensor electrodes also had some issues as they were in the most "active" area when it comes to movements during the study. Because of this some of the data was excluded as it was not useful because of electrodes presumably disconnecting or noise being too overwhelming. It is presumed that these issues largely stemmed from the neck wrinkles created during extension, as the trapezius was relatively still and the sternocleidomastoid is stretched

rather than compressed, they are not affected by wrinkles to the same extent. Because of this sEMG may not be the best choice for measuring muscle activation in the neck extensors during this type of movement.

Something else, which was remarked upon by several drivers, was the lack of an armrest in the reach truck, however this was not done in the interviews but rather when they adjusted their seats. As it was not recorded it is not clear how many commented – but it was significant enough to cause notes being taken on several occasions and of a discussion about adding an armrest being opened by representatives from Toyota Material Handling Europe. This was not done as it could have impacted on the results. Since the armrest was not evaluated it was not asked for when ordering the truck, but it seems as though the drivers who participated were in large part used to it. As this could affect the results it was not added, but it also means that the results may vary if there had been an armrest. As the left side of the body saw a higher activation, as discussed above, an armrest on the left-hand side of the driver may have assisted them in lowering the load.

The pallet used during the study was not a standard EU-pallet – but a metal pallet weighing 500 kilos. This was noted by participants early on as it made it harder to fit the forks in the holes correctly. It may have been better to strap the pallet to an EU-pallet; however, this was a change which would have affected the complexity of the task. This may have affected how the drivers approached sliding the forks in, but the concurrent study also looked at cognitive strain and changed complexity could have seriously impacted it. As it was the complexity may have simulated a higher cognitive strain than using an EU-pallet with a strapped-on weight. It is possible that this served to simulate tasks which require a higher concentration and where it is important for the drivers to properly look at what they are doing. Such situations may also be where the most damaging movements are is essential as risks may be at a high.

Something which served as a real distraction from the tasks were pressing the button, and a lot of participants turned their heads to press it properly, this was handled by the buffer. However, the buffer was not specialized for each participant but rather a mean for all participants – ideally each press of the button would have its own buffer, but this would be extremely time consuming with 704 presses of the button in total. It would have been possible to perhaps create an individual buffer for each participant though, something which would make the results more robust than they are currently. Though, as all participants were not consequential in their pressing of the button this did not seem worthwhile within the scope of the study and it would introduce a code chunk for each file used which would introduce more space for error.

Another possible distraction could have come from the warehouse environment where the data collection took place not being entirely quiet, but the area where the drivers operated was walled off to avoid spectators or visual distractions. However, it is assumed that the driver's normal environment is not entirely quiet and distraction free either, so it was not considered a huge issue. The same goes for the study leader talking to participants to give them instructions throughout the experiment not considered as being too distracting, additionally instructions while driving can be a part of daily work as well. Though there are systems which remove the necessity of having a person give instructions, drivers can still get instructions during their work into the truck. Therefore, it could be argued that they are used to this to some degree.

It should also be mentioned that though the drivers were aware that the tilt function of the reach truck was being evaluated for development purposes. However, even though they were not told exactly what was being evaluated to avoid bias in the interview – they were equipped with

electrodes on their shoulder and neck. It is entirely possible that they understood something about the purpose of the study and the thematic analysis results should be viewed with this in mind. On the other hand, the results of the thematic results align with previous research where this was not the case.

The drivers also had individual driving styles – there was a discussion about asking them to do the task in a specific way however then it would not have emulated their normal working environment. Furthermore, the number of variables to control in their driving style would have been overwhelming and forcing them to keep in mind how to do tasks they are used to would increase their mental load unnecessarily. Thus, the way the drivers lined themselves up, how much they drove the reach truck or pushed only the forks forward when reaching the right height was not controlled. Neither were how ergonomically correct their seat settings were nor how they used the equipment available to them. As all the drivers were used to driving the tilt to some capacity it was decided that letting them act as "normal" was the best way forward to get results which represent their reality.

Of course, the concurrent study also needs to be discussed. The concurrent study included eye-tracking glasses which participants without their own glasses were asked to wear. In the end 15 out of 22 participants did wear the glasses for their entire drive. This could have affected their performance as it was an additional piece of equipment. It was also used by drivers who presumably do not normally wear any glasses which made it a novel experience for them in several ways. This may have led to them having a harder time seeing what they were doing, but none of them mentioned it in their interview. On the contrary improved sight in relation to tilt came up, if the glasses had been a big problem they could reasonably have come up when the topic of sight was discussed.

One participant did have to remove the glasses during the first movement as they had removed their normal glasses, but quickly realized they were needed to complete the work safely. The task was stopped and the glasses removed, then the task was restarted from the first movement.

## 7.3 Future Research

Future research could impact and support the results of the study in several ways. First off, in order to get a better understanding of the results future research should focus on how reliable sEMG on the neck extensors is. As the neck extensors had a high activation it is essential to understand if it is about the activation of the muscles – or if the movements of the neck and accompanying wrinkled affect the results to a point where they are not robust enough to analyze.

Alternative methods for studying neck movements should also be explored. The results of the study show that the tilt setting helps, but not to the extent which participants seem to prefer it. It may be that the muscle activation of larger muscles is not the most bodily stressful part of the movement. Previous research has shown that truck drivers get into awkward positions (Flodin et al., 2018). It may be smaller muscles which are affected. Future studies could map the neck muscles and their activation to a larger extent and then study how truck drivers move – relating it to the muscle activation from the mapping. Furthermore, future studies should explore if other parts of the neck, such as ligaments or bones, are affected by using the tilt cabin.

Both this study and previous work which relates to the thematic results (Túren & Thorslund, 2026) used participants who were used to the Toyota BT Reflex tilt truck. Future work should also investigate how participants who do not use the tilt truck feel about it, preferably in a longer format study to let them experience the difference for a longer period of time.

Reproduction studies and studies including more participants are also a good way to move forward to support or debunk the results. It may also be important to see if the results can be recreated when the researchers and participants are not affiliated with Toyota Material Handling Europe beyond using one of their tilt trucks.

As results also showed higher activation on the left side, particularly in the neck extensors, the use of an armrest should also be further explored. As Hoy et al. (2005) link neck extension to muscle pain it could be crucial to understand how much the armrest can reduce activation on the left side.

It was also noted that the activation was sometimes higher when the drivers were at the eight-meter height, compared to the eleven-meter height. The reason for this could be explored further by studying the difference in how drivers approach the different heights, to gain insights into how the drivers work, which could eventually lead to new or improved ergonomic solutions.

Lastly, this study collected data simultaneously as another study and the results from both studies should be analyzed together to contextualize the results.

# 8. Conclusion

The aim of the study was to deepen the understanding of how the tilt functionality in the Toyota BT Reflex reach truck with a tilting cabin affects drivers, focusing on their neck muscles and reported experience. The study utilized a mixed methods approach and recruited 22 participants. Specifically looking at muscle activation in the neck through surface electromyography (sEMG) and driver attitude with interviews analyzed thematically.

The sEMG results indicated that the tilt function may significantly decrease muscle activation in the neck, potentially lessening strain. This was shown both through analyzing the mean activation using linear mixed models, and through looking at max activation values with barplots. The muscle activation was consequently lessened by using tilt, however not as much as previous research has indicated.

The thematic results showed that drivers have a positive attitude towards, and experience of, the tilt cabin. It also showed improved sightline as a clear theme, something which was put forwarded as positive, though not directly related to a lessened strain on the neck. This is in line with previous work where the positive attitude towards the tilt has also been prevalent. The thematic results also showed that drivers reported feeling less strained in the neck when using tilt, aligning with the sEMG findings.

Both sEMG and thematic results suggests that the tilting cabin is a beneficial ergonomical solution which is appreciated by drivers. The results also support previous work, which in combination with the alignment with each other, supports both the chosen methods and their results. Overall, the study shows that tilt setting is an effective ergonomic tool both by objective and self-reported measures.

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# 10. Appendix

1. Consent form

# Samtyckesblankett

Vad är syftet med studien och vad kommer att ske under den?
Målet med studien är att utvärdera hur ergonomisk Toyotas tiltande truck är för att kunna förbättra den i framtiden. Ditt deltagande bidrar till att kommande versioner av trucken kan bli mer ergonomiska och en djupare förståelse för vad som fungerar och i längden så kan det leda till en bättre arbetsmiljö för truckförare. Under studien så kommer du att få köra en truck i inomhusmiljö och det är viktigt att du har ett giltigt truckkörkort och tillstånd för att kunna delta. Under din körning så kommer du att ha elektroder påsatta på hals och på övre ryggen. Notera att platsen där elektroderna sitter kommer att behöva rakas för att elektroderna ska få bra kontakt med huden.
Du kan eventuellt även få bära ett par eye-tracking glasögon.

Inspelningar
Under tiden som du kör så kommer du att spelas in med kamera som spelar in både bild och ljud. Stillbilder från inspelning samt kortare videoklipp kan komma att användas i kommande rapporter och presentationer men ansikte och eventuella distinkta märken på kläder kommer att suddas ut i anonymiseringssyfte.
Efter att du har genomfört din körning kommer du att delta i en kortare intervju som också spelas in – men endast ljud nyttjas från denna och ingen video kommer att erhållas. Ljudet kan komma att transkriberas och ingen del av transkriptionen som kan leda till din identifikation kommer i det fallet att delas.
Även eye-tracking glasögonen kommer också att spela in hur din blick går under körningen om du bär dem.

Behandling av personliga uppgifter.
Dina personliga uppgifter kommer att hanteras enligt GDPR och kommer inte att sparas mer än ett år efter studiens genomförande. Du kommer att få ett deltagarnummer som kopplas till ditt namn. Det sparas endast lokalt på den studieansvariges dator i syftet att kunna koppla dig till dina resultat och inspelningar om du vill dra tillbaka ditt samtycke. Bilder, videor och ljudinspelningar räknas som personliga uppgifter och behandlas därefter.

Tillbakadragande av samtycke
Du kan närsomhelst under studiens gång avbryta ditt deltagande och få dina uppgifter och resultat raderade. Om du vill dra tillbaka samtycke efter genomförandet kontaktar du ansvarig för studien.
Viktigt att notera är att då dina uppgifter kommer att raderas och publicerade stillbilder och videoklipp kommer att anonymiseras så kommer det ett år efter studien inte längre att gå att koppla dig till videoklippen.

Studiens resultat
Studiens resultat kommer att rapporteras i två vetenskapliga rapporter och kan även komma att publiceras i vetenskapliga journaler eller på konferenser. Om du vill ha tillgång till resultaten kan du alltid kontakta studieansvarig.
Kontakt

Ansvarig för studien är Sofia Davidsson.
Mejl: Sofia.Davidsson@toyota-industries.eu.
Samtycke
Genom att skriva under så intygar du att du har läst och förstått information i samtyckesblanketten och ger ditt samtycke till att delta i studien. Du har även rätt till en kopia av samtyckesblanketten.
Datum och ort

Namnteckning

Namnförtydligande

2. Schedule for Pressing the Button

| Press of button (sequentially numbered) | Action |
|---|---|
| 1 | Before rest |
| 2 | After rest |
| 3 | Before full head movement |
| 4 | After full head movement |
| 5 | Before clenching |
| 6 | After clenching |
| 7 | Before starting the drive |
| 8 | Before lift |
| 9 | Before pallet is lifted 8m |
| 10 | Before pallet is lowered 8m |
| 11 | When pallet is on the floor, before lift, 8m |
| 12 | Before pallet is put back 8m |
| 13 | After pallet is put back 8m |
| 14 | After forks are at floor level |
| 15 | Before lift |
| 16 | Before pallet is lifted 11m |
| 17 | Before pallet is lowered 11m |
| 18 | When pallet is on the floor, before lift, 11m |
| 19 | Before pallet is put back 11m |
| 20 | After pallet is put back 11m |
| 21 | After forks are at floor level |
| Second Round | |
| 22 | Before starting the drive |
| 23 | Before lift |
| 24 | Before pallet is lifted 8m |
| 25 | Before pallet is lowered 8m |
| 26 | When pallet is on the floor, before lift, 8m |
| 27 | Before pallet is put back 8m |
| 28 | After pallet is put back 8m |
| 29 | After forks are at floor level |
| 30 | Before lift |
| 31 | Before pallet is lifted 11m |
| 32 | Before pallet is lowered 11m |
| 33 | When pallet is on the floor, before lift, 11m |
| 34 | Before pallet is put back 11m |
| 35 | After pallet is put back 11m |
| 36 | After forks are at floor level |

3. Interview Questions
Kändes det en skillnad mellan att använda trucken med eller utan tiltande hytt?

Kände du en skillnad i ryggen?

Kände du en skillnad i nacken?

Kände du en skillnad i axlarna?

Brukar du få ont, stelna till eller få andra problem i kroppen när du kör truck och vad är det som händer då?

Ser du något negativt med tilten?

Har du några andra tankar som inte har kommit fram?

4. Full table LMM 1 (Muscle and Setting), from 6.1.1
Full table of the results for the fourth Linear Mixed Model, codes: Sterno = Sternocleidomastoid, Traps = upper trapezius, Without = Without tilt setting. Codes: 0 < *** < 0,001 < ** < 0,01 < * < 0,05 < . < 0.1.

| | Estimate Std. | Error | df | *t*-value | *p*-value |
|---|---|---|---|---|---|
| Not Normalized | | | | | |
| REML: 30385.1 | | | | | |
| (Intercept) | 52.300 | 4.074 | 29.451 | 12.837 | 1.36e-13 *** |
| Without | -2.825 | 2.416 | 2997.073 | -1.169 | 0.2424 |
| Sterno | -25.581 | 2.353 | 3001.514 | -10.874 | < 2e-16 *** |
| Traps | -10.137 | 2.353 | 3001.514 | -4.309 | 1.69e-05 *** |
| Without:sterno | 5.339 | 3.298 | 2997.073 | 1.619 | 0.2056 |
| Without:traps | 6.452 | 3.298 | 2997.073 | 1.956 | 0.0505 . |
| Normalized | | | | | |
| REML: 10560.9 | | | | | |
| (Intercept) | 5.91691 | 0.75715 | 21.88592 | 7.815 | 9.01e-08 *** |
| Without | 0.64814 | 0.17682 | 2087.00000 | 3.666 | 0.000253 *** |
| Traps | -2.72459 | 0.17682 | 2087.00000 | -15.409 | < 2e-16 *** |
| Without:Traps | -0.06705 | 0.25006 | 2087.00000 | -0.268 | 0.788618 |

5. Full table LMM 2 (Side), from 6.1.2
Full table of the results for the second Linear Mixed Model, codes: Sterno = Sternocleidomastoid, Traps = upper trapezius, Right = right side of body, Without = Without tilt setting. Codes: 0 < *** < 0,001 < ** < 0,01 < * < 0,05 < . < 0.1.

| | Estimate Std. | Error | df | *t*-value | *p*-value |
|---|---|---|---|---|---|
| Not Normalized | | | | | |
| REML: 29410.4 | | | | | |
| (Intercept) | 82.9697 | 4.2452 | 34.7489 | 19.544 | < 2e-16 *** |
| Without | -5.0699 | 2.9184 | 2991.0533 | -1.737 | 0.0824 . |
| Right | -61.3556 | 2.9184 | 2991.0533 | -21.024 | < 2e-16 *** |
| Sterno | -53.5918 | 2.8296 | 2992.7587 | -18.940 | < 2e-16 *** |
| Traps | -24.6607 | 2.8296 | 2992.7587 | -8.715 | < 2e-16 *** |
| Without:Right | 4.4902 | 4.1272 | 2991.0533 | 1.088 | 0.2767 |
| Without:Sterno | 5.5001 | 3.9840 | 2991.0533 | 1.381 | 0.1675 |
| Without:Traps | 9.1613 | 3.9840 | 2991.0533 | 2.299 | 0.0215 * |
| Right:Sterno | 56.0380 | 3.9840 | 2991.0533 | 14.066 | < 2e-16 *** |
| Right:Traps | 29.0640 | 3.9840 | 2991.0533 | 7.295 | 3.8e-13 *** |
| Without:Right:Sterno | -0.3225 | 5.6343 | 2991.0533 | -0.057 | 0.9544 |
| Without:Right:Traps | -5.4185 | 5.6343 | 2991.0533 | -0.962 | 0.3363 |
| Normalized | | | | | |
| REML: 10408.3 | | | | | |
| (Intercept) | 6.0727 | 0.7661 | 22.9443 | 7.926 | 5.11e-08 *** |
| Without | 0.4523 | 0.2411 | 2083.0000 | 1.876 | 0.0608 . |
| Right | -0.3116 | 0.2411 | 2083.0000 | -1.293 | 0.1963 |

| | | | | | |
|---|---|---|---|---|---|
| Traps | -1.8123 | 0.2411 | 2083.0000 | -21.024 | 8.25e-14 *** |
| Without:Right | 0.3916 | 0.3409 | 2083.0000 | 1.149 | 0.2508 |
| Without:Traps | 0.1527 | 0.3409 | 2083.0000 | 0.448 | 0.6582 |
| Right:Traps | -1.8246 | 0.3409 | 2083.0000 | -18.940 | 9.66e-08 *** |
| Without:Right:Traps | -0.4396 | 0.4822 | 2083.0000 | -0.912 | 0.3620 |

6. Full table LMM 3 (Task), from 6.1.3

Full table of the results for the third Linear Mixed Model, codes: Sterno = Sternocleidomastoid, Traps = upper trapezius, Right = right side of body, Without = Without tilt setting, Work = Work with pallet at height, Down = Moving pallet down. Codes: 0 < *** < 0,001 < ** < 0,01 < * < 0,05 < . < 0.1.

| | Estimate | Std. Error | df | *t*-value | *p*-value |
|---|---|---|---|---|---|
| Not Normalized | | | | | |
| REML : 30298 | | | | | |
| (Intercept) | 53.4865 | 4.7303 | 53.4003 | 11.307 | 8.67e-16 *** |
| Without | -4.1468 | 4.1699 | 2985.0721 | -0.994 | 0.3201 |
| Sterno | -26.1982 | 4.0369 | 2986.6111 | -6.490 | 1.00e-10 *** |
| Traps | -8.9838 | 4.0369 | 2986.6111 | -2.225 | 0.0261 * |
| Work | 6.6657 | 4.1699 | 2985.0721 | 1.599 | 0.1100 |
| Down | -10.2243 | 4.1699 | 2985.0721 | -2.452 | 0.0143 * |
| Without:Sterno | 6.8533 | 5.6925 | 2985.0721 | 1.204 | 0.2287 |
| Without:Traps | 9.0356 | 5.6925 | 2985.0721 | 1.587 | 0.1126 |
| Without:Work | -0.1690 | 5.8971 | 2985.0721 | -0.029 | 0.9771 |
| Without:Down | 4.1349 | 5.8971 | 2985.0721 | 0.701 | 0.4832 |
| Sterno:Work | -8.1843 | 5.6925 | 2985.0721 | -1.438 | 0.1506 |
| Traps:Work | -8.5633 | 5.6925 | 2985.0721 | -1.504 | 0.1326 |
| Sterno:Down | 10.0353 | 5.6925 | 2985.0721 | 1.763 | 0.0780 . |
| Traps:Down | 5.1036 | 5.6925 | 2985.0721 | 0.897 | 0.3700 |
| Without:Sterno:Work | 2.3792 | 8.0504 | 2985.0721 | 0.296 | 0.7676 |
| Without:Traps:Work | -0.4131 | 8.0504 | 2985.0721 | -0.051 | 0.9591 |
| Without:Sterno:Down | -6.9223 | 8.0504 | 2985.0721 | -0.860 | 0.3899 |
| Without:Traps:Down | -7.3378 | 8.0504 | 2985.0721 | -0.911 | 0.3621 |
| Normalized | | | | | |
| REML: 10551.9 | | | | | |
| (Intercept) | 5.98558 | 0.77744 | 24.32610 | 7.699 | 5.64e-08 *** |
| Without | 0.75724 | 0.30578 | 2079.00000 | 2.476 | 0.0133 * |
| Traps | -2.66599 | 0.30578 | 2079.00000 | -8.719 | < 2e-16 *** |
| Work | -0.40085 | 0.30578 | 2079.00000 | -1.311 | 0.1900 |
| Down | 0.19486 | 0.30578 | 2079.00000 | 0.637 | 0.5240 |

| | | | | | |
|---|---|---|---|---|---|
| Without:Traps | -0.01239 | 0.43244 | 2079.00000 | -0.029 | 0.9772 |
| Without:Work | 0.48029 | 0.43244 | 2079.00000 | 1.111 | 0.2668 |
| Without:Down | -0.80760 | 0.43244 | 2079.00000 | -1.868 | 0.0620 . |
| Traps:Work | 0.30685 | 0.43244 | 2079.00000 | 0.710 | 0.4780 |
| Traps:Down | -0.48264 | 0.43244 | 2079.00000 | -1.116 | 0.2645 |
| Without:Traps:Work | -0.69003 | 0.61156 | 2079.00000 | -1.128 | 0.2593 |
| Without:Traps:down | 0.52604 | 0.61156 | 2079.00000 | 0.860 | 0.3898 |

7. Full table LMM 4 (Initial Setting), from 6.1.4

Full table of the results for the fourth Linear Mixed Model, codes: Sterno = Sternocleidomastoid, Traps = upper trapezius, Without = Without tilt setting, Start Without = Started without the tilt setting. Codes: 0 < *** < 0,001 < ** < 0,01 < * < 0,05 < . < 0.1

| | Estimate Std. | Error | df | *t*-value | *p*-value |
|---|---|---|---|---|---|
| Not Normalized | | | | | |
| REML: 30338.4 | | | | | |
| (Intercept) | 60.291 | 5.672 | 29.549 | 10.629 | 1.30e-11 *** |
| Without | -7.535 | 3.503 | 2992.069 | -2.151 | 0.03157 * |
| Sterno | -33.354 | 3.333 | 2999.854 | -10.009 | < 2e-16 *** |
| Traps | -16.613 | 3.333 | 2999.854 | -4.985 | 6.55e-07 *** |
| Start Without | -15.250 | 8.277 | 27.696 | -1.842 | 0.07614 . |
| Without:Sterno | 8.358 | 4.634 | 2992.069 | 1.804 | 0.07140 . |
| Without:Traps | 10.313 | 4.634 | 2992.069 | 2.225 | 0.02613 * |
| Without:Start Without | 8.949 | 4.829 | 2992.069 | 1.853 | 0.06396 . |
| Sterno:Start Without | 14.770 | 4.706 | 2996.208 | 3.138 | 0.00172 ** |
| Traps:Start Without | 11.917 | 4.706 | 2996.208 | 2.532 | 0.01139 * |
| Without:Sterno:Start Without | -5.230 | 6.601 | 2992.069 | -0.792 | 0.42822 |
| Without:Traps:Start Without | -7.081 | 6.601 | 2992.069 | -1.073 | 0.28347 |
| Normalized | | | | | |
| REML: 10547.8 | | | | | |
| (Intercept) | 6.0868 | 1.0491 | 29.549 | 5.802 | 9.62e-06 *** |
| Without | 0.2777 | 0.2389 | 2084.0000 | 1.162 | 0.2452 |
| Traps | -2.5853 | 0.2389 | 2084.0000 | -10.821 | < 2e-16 *** |
| Start Without | -37.38 | 1.5560 | 20.8011 | -0.240 | 0.8125 |
| Without:Traps | 0.2290 | 0.3379 | 2084.0000 | 0.678 | 0.4980 |
| Without:Start Without | 0.8149 | 0.3544 | 2084.0000 | 2.300 | 0.0216 * |
| Traps:Start Without | -0.3065 | 0.3544 | 2084.0000 | -0.865 | 0.3872 |
| Without:Traps:Start Without | -0.6513 | 0.5011 | 2084.0000 | -1.300 | 0.1939 |

8. Complete results mean difference between heights

*Summary of Activation Difference Between Heights, Normalized Data*

| Variable | Movement | With | Without | Without - With |
|---|---|---|---|---|
| Trapezius Left | Down | 0.473775872 | 1.008403946 | 0.534628074 |
| Trapezius Left | Down second | 0.349247838 | 0.345088860 | -0.004158978 |
| Trapezius Left | Work | 0.252436742 | -0.183046481 | -0.069390261 |
| Trapezius Left | Work second | 1.113428856 | 0.395785485 | -0.717643371 |
| Trapezius Left | Up | 0.650727510 | 1.781616872 | 1.130889363 |
| Trapezius Left | Up second | 1.473479107 | 0.202069629 | -1.271409479 |
| Trapezius Right | Down | 0.130814128 | 0.461865216 | 0.331051088 |
| Trapezius Right | Down second | -0.628159185 | -0.001424615 | -0.626734570 |
| Trapezius Right | Work | -0.006471473 | -0.096146895 | 0.089675422 |
| Trapezius Right | Work second | 0.080220858 | -0.345132018 | 0.264911160 |
| Trapezius Right | Up | -0.114053470 | 0.385839315 | 0.271785845 |
| Trapezius Right | Up second | 0.188382812 | 0.452028177 | 0.263645365 |
| Sternocleidomastoid Left | Down | 0.518684004 | 0.825191546 | 0.306507542 |
| Sternocleidomastoid Left | Down second | -0.385281175 | -0.035455231 | -0.349825944 |
| Sternocleidomastoid Left | Work | 1.326252390 | 0.120738180 | -1.205514210 |
| Sternocleidomastoid Left | Work second | 1.237153505 | -0.291075348 | -0.946078157 |
| Sternocleidomastoid Left | Up | 0.678461134 | 1.463166178 | 0.784705043 |
| Sternocleidomastoid Left | Up second | 0.788130156 | 0.265575877 | -0.522554279 |
| Sternocleidomastoid Right | Down | 0.363554301 | 1.322332790 | 0.958778489 |
| Sternocleidomastoid Right | Down second | -1.199498529 | 0.374476166 | -0.825022364 |
| Sternocleidomastoid Right | Work | 0.561080638 | 0.393212589 | -0.167868049 |
| Sternocleidomastoid Right | Work second | -0.085326071 | 0.052694216 | -0.032631855 |
| Sternocleidomastoid Right | Up | 1.502328952 | 1.693111126 | 0.190782174 |
| Sternocleidomastoid Right | Up second | 0.662230687 | 1.452145993 | 0.789915305 |
| Neck Extensor Left | Down | 1.451142573 | 4.042139290 | 2.590996717 |
| Neck Extensor Left | Down second | 2.040085928 | 2.230256311 | 0.190170383 |
| Neck Extensor Left | Work | 4.656488920 | 2.680084374 | -1.976404545 |
| Neck Extensor Left | Work second | 3.156754832 | 2.617755429 | -0.538999403 |
| Neck Extensor Left | Up | 6.504792722 | 7.336892917 | 0.832100195 |
| Neck Extensor Left | Up second | 4.472907838 | 5.931992486 | 1.459084648 |
| Neck Extensor Right | Down | 0.504347841 | 0.586424852 | 0.082077011 |
| Neck Extensor Right | Down second | -2.586069659 | -1.837712743 | -0.748356917 |
| Neck Extensor Right | Work | -0.880582875 | -0.743474193 | -0.137108682 |
| Neck Extensor Right | Work second | 0.077086649 | -2.626655952 | 2.549569303 |
| Neck Extensor Right | Up | 0.953871947 | 3.350798204 | 2.396926257 |
| Neck Extensor Right | Up second | 0.632685974 | 0.606554389 | -0.026131585 |

*Summary of Activation Difference Between Heights, Normalized Data*

| Variable | Movement | With | Without | Without - With |
|---|---|---|---|---|
| Trapezius Left | Down | 0.473775872 | 1.008403946 | 0.534628074 |
| Trapezius Left | Down second | 0.349247838 | 0.345088860 | -0.004158978 |
| Trapezius Left | Work | 0.252436742 | -0.183046481 | -0.069390261 |
| Trapezius Left | Work second | 1.113428856 | 0.395785485 | -0.717643371 |
| Trapezius Left | Up | 0.650727510 | 1.781616872 | 1.130889363 |
| Trapezius Left | Up second | 1.473479107 | 0.202069629 | -1.271409479 |
| Trapezius Right | Down | 0.130814128 | 0.461865216 | 0.331051088 |
| Trapezius Right | Down second | -0.628159185 | -0.001424615 | -0.626734570 |
| Trapezius Right | Work | -0.006471473 | -0.096146895 | 0.089675422 |

| Variable | Movement | With | Without | Without - With |
|---|---|---|---|---|
| Trapezius Right | Work second | 0.080220858 | -0.345132018 | 0.264911160 |
| Trapezius Right | Up | -0.114053470 | 0.385839315 | 0.271785845 |
| Trapezius Right | Up second | 0.188382812 | 0.452028177 | 0.263645365 |
| Sternocleidomastoid Left | Down | 0.518684004 | 0.825191546 | 0.306507542 |
| Sternocleidomastoid Left | Down second | -0.385281175 | -0.035455231 | -0.349825944 |
| Sternocleidomastoid Left | Work | 1.326252390 | 0.120738180 | -1.205514210 |
| Sternocleidomastoid Left | Work second | 1.237153505 | -0.291075348 | -0.946078157 |
| Sternocleidomastoid Left | Up | 0.678461134 | 1.463166178 | 0.784705043 |
| Sternocleidomastoid Left | Up second | 0.788130156 | 0.265575877 | -0.522554279 |
| Sternocleidomastoid Right | Down | 0.363554301 | 1.322332790 | 0.958778489 |
| Sternocleidomastoid Right | Down second | -1.199498529 | 0.374476166 | -0.825022364 |
| Sternocleidomastoid Right | Work | 0.561080638 | 0.393212589 | -0.167868049 |
| Sternocleidomastoid Right | Work second | -0.085326071 | 0.052694216 | -0.032631855 |
| Sternocleidomastoid Right | Up | 1.502328952 | 1.693111126 | 0.190782174 |
| Sternocleidomastoid Right | Up second | 0.662230687 | 1.452145993 | 0.789915305 |

9. Translated Interview Quotes
Inte ifylld än utan det är bara översättningar av de utdrag som använts.

**Excerpt 1:**
English: “I thought somehow it felt much better in one way to drive the one where you don’t tilt with it, I felt more in the neck and all of that.” (P18)

Swedish: “Jag tyckte ju på ett sätt att det kändes mycket bättre på ett sätt att köra den man inte tiltar med, jag kände ju mer efter i nacken o allt sånt där.” (18)

**Excerpt 2**
English: ”a little in neck and in shoulder but ah, it usually goes away as the week passes (I:mm) but course’ it’s always there but it is not something that’s straining so. But as I said I think you need to keep moving a little somehow to work at it otherwise it is probably hard a- then you can get stiff, I would say at least”(9)

Swedish: ”lite i nacken och i axeln men ah det brukar försvinna efter veckans gång liksom (I: mm) men de tär klart det finns ju alltid där men det är inte nått som är påfrestande så. Men som sagt jag tror man nog måste hålla igång lite på nått sätt för att bearbeta det annars är det nog ganska svårt o - då kan man nog bli ganska stel, skulle jag säga iallafall” (9)

**Excerpt 3**
English: “Yeah of course a bit at the top it is better leaned back you see the sides and edges a little better.” (P6)

Swedish: Ja klart men lite längst upp det är bättre man ser lutade bakåt ser lite bättre sidorna och kanterna.” (P6)

**Excerpt 4**
English: “Big difference (I:mm) *unclear* leaning you can kind of see the edges.” (P3)

Swedish: ”Stor skillnad (I:mm) € att vara lutad man kan se kanterna typ.” (P3)

**Excerpt 5**
English: “You get more ehhm of a handle ha no but it feels safer when you don’t have to sit so bent with the neck or something…” (P12)

Swedish: ”Man får mer eehm koll ha nej men det känns tryggare när man inte behöver sitta så böjd med nacken eller nåt…”